



Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

- VERIFIED 0.80 - 1.00** 3+ independent sources including media
- CONFIRMED 0.50 - 0.79** 2 sources or 1 high-quality media
- EMERGING 0.25 - 0.49** Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	303	Media
TechCrunch AI	7	Media
MIT Tech Review	4	Media
The Verge AI	3	Media
HackerNews	2	Community

VERIFIED INTELLIGENCE — 1 story

VERI

Positive Alignment: Artificial Intelligence for Human Flourishing

ArXiv CS.AI +1 source

80%

arXiv:2605.10310v1 Announce Type: new Abstract: Existing alignment research is dominated by concerns about safety and preventing harm: safeguards, controllability, and compliance. This paradigm of alignment parallels early psychology's focus on mental illness: necessary but...

<https://arxiv.org/abs/2605.10310>

CONFIRMED SIGNALS — 317 stories

CONF

Show HN: AI Agents in 30 Lines of YAML: Lowdefy v5.3

HackerNews 50% HN 5

No summary — see link for full article.

<https://lowdefy.com/articles/lowdefy-agents/>

CONF

Show HN: ChatGPT Exporter - Local DOM to Word/PDF Parser

HackerNews 50% HN 5

No summary — see link for full article.

<https://chromewebstore.google.com/detail/chatgpt-exporter-save-cha/ploaaddkflkapjfbfapm...>

CONF

Thinking Machines wants to build an AI that actually listens while it talks

TechCrunch AI 50%

Right now, every AI model you've ever used works the same way. You talk, it listens. It responds, you listen. Thinking Machines is trying to change that by building a model that processes your input and generates a response at the same time, so it's more like a phone call than a...

<https://techcrunch.com/2026/05/11/thinking-machines-wants-to-build-an-ai-that-actually-...>

CONF

Riding an AI rally, Robinhood preps second retail venture IPO

TechCrunch AI 50%

Robinhood files confidentially for its second venture fund, this time targeting growth and early-stage startups.

<https://techcrunch.com/2026/05/11/riding-an-ai-rally-robinhood-preps-second-retail-vent...>

CONF

GM just laid off hundreds of IT workers to hire those with stronger AI skills

TechCrunch AI 50%

Some of the positions focus on AI-native development, data engineering and analytics, cloud-based engineering, and agent and model development as well as prompt engineering and new AI workflows.

<https://techcrunch.com/2026/05/11/gm-just-laid-off-hundreds-of-it-workers-to-hire-those...>

CONF

Digg tries again, this time as an AI news aggregator

TechCrunch AI 50%

In an email to beta testers, the company said the site's goal is to "track the most influential voices in a space" and to surface the news that's actually worth "paying attention to."

<https://techcrunch.com/2026/05/11/digg-tries-again-this-time-as-an-ai-news-aggregator/>

CONF

There aren't enough rockets for space data centers — Cowboy Space raised \$275M to build them

TechCrunch AI 50%

The apparently insatiable demand for AI compute has data center entrepreneurs looking to the stars. There's a key problem: There aren't enough rockets to put data centers in orbit around Earth, and they're too expensive.

<https://techcrunch.com/2026/05/11/there-arent-enough-rockets-for-space-data-centers-cow...>

CONF

Voice AI in India is hard — Wispr Flow is betting on it anyway

TechCrunch AI 50%

Wispr Flow says growth accelerated in India after its Hinglish rollout, even as voice AI products continue to face challenges.

<https://techcrunch.com/2026/05/09/voice-ai-in-india-is-hard-wispr-flow-is-betting-on-it...>

0 CONF

Three things in AI to watch, according to a Nobel-winning economist

MIT Tech Review 50%

This story originally appeared in The Algorithm, our weekly newsletter on AI. To get stories like this in your inbox first, sign up here. A few months before he was awarded the Nobel Prize in economics in 2024, Daron Acemoglu published a paper that earned him few fans in Silicon...

<https://www.technologyreview.com/2026/05/11/1137090/three-things-in-ai-to-watch-accordi...>

1 CONF

Fostering breakthrough AI innovation through customer-back engineering

MIT Tech Review 50%

Despite years of digitization, organizations capture less than one-third of the value expected from digital investments, according to McKinsey research. That's because most big companies begin with technological capabilities and bolt applications onto them, rather than starting...

<https://www.technologyreview.com/2026/05/11/1136967/fostering-breakthrough-ai-innovatio...>

2 CONF

Implementing advanced AI technologies in finance

MIT Tech Review 50%

In finance departments that have long been defined by precision and control, AI has arrived less as a neatly managed upgrade than as a quiet insurgency. Employees are already using it while leadership races to impose structure, governance, and strategy after the fact. The result...

<https://www.technologyreview.com/2026/05/11/1136786/implementing-advanced-ai-technologi...>

3 CONF

The Download: AI malaise and babymaking tech

MIT Tech Review 50%

This is today's edition of The Download, our weekday newsletter that provides a daily dose of what's going on in the world of technology. We've entered the era of AI malaise AI is spreading everywhere, and it is not going away. But what will it do? What effect will it have on...

<https://www.technologyreview.com/2026/05/08/1136985/the-download-ai-malaise-babymaking-...>

4 CONF

OpenAI just released its answer to Claude Mythos

The Verge AI 50%

OpenAI is launching Daybreak, an AI initiative focused on detecting and patching vulnerabilities before attackers find them. Daybreak uses the Codex Security AI agent that launched in March to create a threat model based on an organization's code and focus on possible attack...

<https://www.theverge.com/ai-artificial-intelligence/928342/openai-daybreak-security-ai>

5 CONF

Here's what Mira Murati's AI company is up to

The Verge AI 50%

Thinking Machines, the AI company founded by former OpenAI CTO Mira Murati, announced Monday that it's working on something called "interaction models." The idea behind interaction models, according to Thinking Machines, is that they will let people "collaborate with AI the way..."

<https://www.theverge.com/ai-artificial-intelligence/928309/mira-murati-thinking-machine...>

6 CONF

Google stopped a zero-day hack that it says was developed with AI

The Verge AI 50%

For the first time, Google says it has spotted and stopped a zero-day exploit developed with AI. According to a report from Google Threat Intelligence Group (GTIG), "prominent cyber crime threat actors" were planning to use the vulnerability for a "mass exploitation event" that...

<https://www.theverge.com/tech/928007/google-ai-zero-day-exploit-stopped>

7 CONF

Where Reliability Lives in Vision-Language Models: A Mechanistic Study of Attention, Hidden States, and Causal Circuits

ArXiv CS.AI 50%

arXiv:2605.08200v1 Announce Type: new Abstract: A pervasive intuition holds that vision-language models (VLMs) are most trustworthy when their attention maps look sharp: concentrated attention on the queried region should imply a confident, calibrated answer. We test this...

<https://arxiv.org/abs/2605.08200>

8 CONF

Spatial Priming Outperforms Semantic Prompting: A Grid-Based Approach to Improving LLM Accuracy on Chart Data Extraction

ArXiv CS.AI 50%

arXiv:2605.08220v1 Announce Type: new Abstract: The automated extraction of data from scientific charts is a critical task for large-scale literature analysis. While multimodal Large Language Models (LLMs) show promise, their accuracy on non-standardized charts remains a...

<https://arxiv.org/abs/2605.08220>

9 CONF

A Unified Pair-GRPO Family: From Implicit to Explicit Preference Constraints for Stable and General RL Alignment

ArXiv CS.AI 50%

arXiv:2605.06375v1 Announce Type: cross Abstract: Large language model (LLM) alignment via reinforcement learning from human preferences (RLHF) suffers from unstable policy updates, ambiguous gradient directions, poor interpretability, and high gradient variance in mainstream...

<https://arxiv.org/abs/2605.06375>

0 CONF

Embeddings for Preferences, Not Semantics

ArXiv CS.AI 50%

arXiv:2605.08360v1 Announce Type: new Abstract: Modern AI is opening the door to collective decision-making in which participants express their views as free-form text rather than voting on a fixed set of candidates. A natural idea is to embed these opinions in a vector space so...

<https://arxiv.org/abs/2605.08360>

1 **CONF****On Distinguishing Capability Elicitation from Capability Creation in Post-Training: A Free-Energy Perspective**

ArXiv CS.AI 50%

arXiv:2605.08368v1 Announce Type: new Abstract: Debates about large language model post-training often treat supervised fine-tuning (SFT) as imitation and reinforcement learning (RL) as discovery. But this distinction is too coarse. What matters is whether a training procedure...

<https://arxiv.org/abs/2605.08368>

2 **CONF****DARE: Diffusion Language Model Activation Reuse for Efficient Inference**

ArXiv CS.AI 50%

arXiv:2605.08134v1 Announce Type: cross Abstract: Diffusion Large Language Models (dLLMs) have emerged as a promising alternative to auto-regressive (AR) models, offering greater expressive capacity and potential for parallel generation and faster inference. However, open-source...

<https://arxiv.org/abs/2605.08134>

3 **CONF****StereoTales: A Multilingual Framework for Open-Ended Stereotype Discovery in LLMs**

ArXiv CS.AI 50%

arXiv:2605.10442v1 Announce Type: cross Abstract: Multilingual studies of social bias in open-ended LLM generation remain limited: most existing benchmarks are English-centric, template-based, or restricted to recognizing pre-specified stereotypes. We introduce StereoTales, a...

<https://arxiv.org/abs/2605.10442>

4 **CONF****Playing games with knowledge: AI-Induced delusions need game theoretic interventions**

ArXiv CS.AI 50%

arXiv:2605.08409v1 Announce Type: new Abstract: Conversational AI has a fundamental flaw as a knowledge interface: sycophantic chatbots induce epistemic entrenchment and delusional belief spirals even in rational agents. We propose the problem does not stem from the AI model,...

<https://arxiv.org/abs/2605.08409>

5 **CONF****A Game Theoretic Free Energy Analysis of Higher Order Synergy in Attention Heads of Large Language Models**

ArXiv CS.AI 50%

arXiv:2605.09515v1 Announce Type: new Abstract: Large language models rely on multihead attention, but interactions among heads remain poorly understood. We apply the Game Theoretic Free Energy Principle (GTfEP): a framework casting multiagent systems as distributed variational...

<https://arxiv.org/abs/2605.09515>

6 **CONF****Alignment as Jurisprudence**

ArXiv CS.AI 50%

arXiv:2605.08416v1 Announce Type: new Abstract: Jurisprudence, the study of how judges should properly decide cases, and alignment, the science of getting AI models to conform to human values, share a fundamental structure. These seemingly distant fields both seek to predict and...

<https://arxiv.org/abs/2605.08416>

7 CONF

MDGYM: Benchmarking AI Agents on Molecular Simulations

ArXiv CS.AI 50%

arXiv:2605.08941v1 Announce Type: new Abstract: The promise of AI-driven scientific discovery hinges on whether AI agents can autonomously design and execute the computational workflows that underpin modern science. Molecular dynamics (MD) simulation presents a natural test bed...

<https://arxiv.org/abs/2605.08941>

8 CONF

TinySSL: Distilled Self-Supervised Pretraining for Sub-Megabyte MCU Models

ArXiv CS.AI 50%

arXiv:2605.08241v1 Announce Type: cross Abstract: Self-supervised learning (SSL) has transformed representation learning for large models, yet remains unexplored for microcontroller (MCU)-class models with fewer than 500K parameters. We identify three obstacles at this scale --...

<https://arxiv.org/abs/2605.08241>

9 CONF

Behavioral Determinants of Deployed AI Agents in Social Networks: A Multi-Factor Study of Personality, Model, and Guardrail Specification

ArXiv CS.AI 50%

arXiv:2605.08463v1 Announce Type: new Abstract: Autonomous AI agents are increasingly deployed in open social environments, yet the relationship between their configuration specifications and their emergent social behavior remains poorly understood. We present a controlled,...

<https://arxiv.org/abs/2605.08463>

0 CONF

AHD Agent: Agentic Reinforcement Learning for Automatic Heuristic Design

ArXiv CS.AI 50%

arXiv:2605.08756v1 Announce Type: new Abstract: Automatic heuristic design (AHD) has emerged as a promising paradigm for solving NP-hard combinatorial optimization problems (COPs). Recent works show that large language models (LLMs), when integrated into well-designed frameworks...

<https://arxiv.org/abs/2605.08756>

1 CONF

AI-Care: A Conversational Agentic System for Task Coordination in Alzheimer's Disease Care

ArXiv CS.AI 50%

arXiv:2605.08480v1 Announce Type: new Abstract: Individuals with Alzheimer's disease (AD) and Alzheimer's disease-related dementia (ADRD) experience memory and thinking changes that impact their ability to use digital daily management tools. For example, adding an event to a...

<https://arxiv.org/abs/2605.08480>

2 CONF

CAMAL: Improving Attention Alignment and Faithfulness with Segmentation Masks

ArXiv CS.AI 50%

arXiv:2605.08325v1 Announce Type: cross Abstract: Many vision datasets now provide segmentation masks in addition to annotated images to support a wide range of tasks. In this work, we propose Class Activation Map Attention Learning (CAMAL), an efficient and scalable method that...

<https://arxiv.org/abs/2605.08325>

3 CONF

OracleTSC: Oracle-Informed Reward Hurdle and Uncertainty Regularization for Traffic Signal Control

ArXiv CS.AI 50%

arXiv:2605.08516v1 Announce Type: new Abstract: Transparent decision-making is essential for traffic signal control (TSC) systems to earn public trust. However, traditional reinforcement learning-based TSC methods function as black boxes with limited interpretability. Although...

<https://arxiv.org/abs/2605.08516>

4 CONF

Human-LLM Dialogue Improves Diagnostic Accuracy in Emergency Care

ArXiv CS.AI 50%

arXiv:2605.08533v1 Announce Type: new Abstract: Clinical decision-making in emergency medicine demands rapid, accurate diagnoses under uncertainty. Despite benchmark progress, evidence for LLMs as interactive aids in live physician workflows remains sparse. MedSyn lets...

<https://arxiv.org/abs/2605.08533>

5 CONF

NaiAD: Initiate Data-Driven Research for LLM Advertising

ArXiv CS.AI 50%

arXiv:2605.09918v1 Announce Type: cross Abstract: Reconciling platform revenue with user experience in LLM advertising motivates a data-centric foundation. We introduce NaiAD, the first comprehensive dataset for LLM-native advertising comprising 58,999 carefully constructed...

<https://arxiv.org/abs/2605.09918>

6 CONF

Agent-X: Full Pipeline Acceleration of On-device AI Agents

ArXiv CS.AI 50%

arXiv:2605.10380v1 Announce Type: new Abstract: LLM-based agents deliver state-of-the-art performance across tasks but incur high end-to-end latency on edge devices. We introduce Agent-X, a software-only, accuracy-preserving framework that accelerates both the prefill and decode...

<https://arxiv.org/abs/2605.10380>

7 CONF

Evaluating Developmental Cognition Capabilities of LLMs

ArXiv CS.AI 50%

arXiv:2605.08549v1 Announce Type: new Abstract: Conversational AI is increasingly personalized around users' preferences, histories, goals, and knowledge, but much less around how users interpret and take up model outputs to construct and understand their reality. We draw on...

<https://arxiv.org/abs/2605.08549>

8 CONF

PRISM: Generation-Time Detection and Mitigation of Secret Leakage in Multi-Agent LLM Pipelines

ArXiv CS.AI 50%

arXiv:2605.10614v1 Announce Type: new Abstract: Multi-agent LLM systems introduce a security risk in which sensitive information accessed by one agent can propagate through shared context and reappear in downstream outputs, even without explicit adversarial intent. We formalise...

<https://arxiv.org/abs/2605.10614>

9 **CONF****Causal Dimensionality of Transformer Representations: Measurement, Scaling, and Layer Structure****ArXiv CS.AI** **50%**

arXiv:2605.08740v1 Announce Type: cross Abstract: Sparse autoencoders (SAEs) decompose transformer residual streams into interpretable feature dictionaries, yet the relationship between SAE width and causal influence on model output has not been systematically characterised. We...

<https://arxiv.org/abs/2605.08740>

0 **CONF****Emergent Semantic Role Understanding in Language Models****ArXiv CS.AI** **50%**

arXiv:2605.09187v1 Announce Type: new Abstract: Understanding how linguistic structure emerges in language models is central to interpreting what these systems learn from data and how much supervision they truly require. In particular, semantic role understanding ("who did what...

<https://arxiv.org/abs/2605.09187>

1 **CONF****Generalization Bounds of Emergent Communications for Agentic AI Networking****ArXiv CS.AI** **50%**

arXiv:2605.08613v1 Announce Type: new Abstract: The evolution of 6G networking toward agentic AI networking (AgentNet) systems requires a shift from traditional data pipelines to task-aware, agentic AI-native communication solutions. Emergent communication, a novel communication...

<https://arxiv.org/abs/2605.08613>

2 **CONF****DiagnosticIQ: A Benchmark for LLM-Based Industrial Maintenance Action Recommendation from Symbolic Rules****ArXiv CS.AI** **50%**

arXiv:2605.08614v1 Announce Type: new Abstract: Monitoring complex industrial assets relies on engineer-authored symbolic rules that trigger based on sensor conditions and prompt technicians to perform corrective actions. The bottleneck is not detection but response: translating...

<https://arxiv.org/abs/2605.08614>

3 **CONF****Iterative Critique-and-Routing Controller for Multi-Agent Systems with Heterogeneous LLMs****ArXiv CS.AI** **50%**

arXiv:2605.08686v1 Announce Type: new Abstract: Multi-agent large language model (LLM) systems often rely on a controller to coordinate a pool of heterogeneous models, yet existing controllers are typically limited to one-shot routing: they select a model once and return its...

<https://arxiv.org/abs/2605.08686>

4 **CONF****Metis: Learning to Jailbreak LLMs via Self-Evolving Metacognitive Policy Optimization****ArXiv CS.AI** **50%**

arXiv:2605.10067v1 Announce Type: cross Abstract: Red teaming is critical for uncovering vulnerabilities in Large Language Models (LLMs). While automated methods have improved scalability, existing approaches often rely on static heuristics or stochastic search, rendering them...

<https://arxiv.org/abs/2605.10067>

5 CONF

When Can Human-AI Teams Outperform Individuals? Tight Bounds with Impossibility Guarantees

ArXiv CS.AI 50%

arXiv:2605.08710v1 Announce Type: new Abstract: Human-AI teams fail to outperform their best member in 70% of studies, yet no theory specifies when complementarity is achievable. We derive tight bounds for the broad class of confidence-based aggregation rules by integrating...

<https://arxiv.org/abs/2605.08710>

6 CONF

Bias by Necessity: Impossibility Theorems for Sequential Processing with Convergent AI and Human Validation

ArXiv CS.AI 50%

arXiv:2605.08716v1 Announce Type: new Abstract: Are certain cognitive biases mathematically inevitable consequences of sequential information processing? We prove that primacy effects, anchoring, and order-dependence are architecturally necessary in autoregressive language...

<https://arxiv.org/abs/2605.08716>

7 CONF

Value-Decomposed Reinforcement Learning Framework for Taxiway Routing with Hierarchical Conflict-Aware Observations

ArXiv CS.AI 50%

arXiv:2605.08754v1 Announce Type: new Abstract: Taxiway routing and on-surface conflict avoidance are coupled safety-critical decision problems in airport surface operations. Existing planning and optimization methods are often limited by online computational cost, while...

<https://arxiv.org/abs/2605.08754>

8 CONF

From Holo Pockets to Electron Density: GPT-style Drug Design with Density

ArXiv CS.AI 50%

arXiv:2605.08767v1 Announce Type: new Abstract: Recent advances in generative modeling have enabled significant progress in structure-based drug design (SBDD). Existing methods typically condition molecule generation on empty binding pockets from holo complexes, overlooking...

<https://arxiv.org/abs/2605.08767>

9 CONF

PATRA: Pattern-Aware Alignment and Balanced Reasoning for Time Series Question Answering

ArXiv CS.AI 50%

arXiv:2602.23161v2 Announce Type: replace Abstract: Time series reasoning demands both the perception of complex dynamics and logical depth. However, existing LLM-based approaches exhibit two limitations: they often treat time series merely as text or images, failing to capture...

<https://arxiv.org/abs/2602.23161>

0 CONF

Mental Health AI Safety Claims Must Preserve Temporal Evidence

ArXiv CS.AI 50%

arXiv:2605.08827v1 Announce Type: new Abstract: The safety of mental health AI is often judged at the wrong temporal scale. Current evaluations typically score isolated responses, endpoint outcomes, or aggregate dialogue quality, while clinically consequential failures may arise...

<https://arxiv.org/abs/2605.08827>

1 CONF

When Agents Overtrust Environmental Evidence: An Extensible Agentic Framework for Benchmarking Evidence-Grounding Defects in LLM Agents

ArXiv CS.AI 50%

arXiv:2605.08828v1 Announce Type: new Abstract: Large language model agents increasingly operate through environment-facing scaffolds that expose files, web pages, APIs, and logs. These observations influence tool use, state tracking, and action sequencing, yet their reliability...

<https://arxiv.org/abs/2605.08828>

2 CONF

GraphReAct: Reasoning and Acting for Multi-step Graph Inference

ArXiv CS.AI 50%

arXiv:2605.07357v2 Announce Type: replace Abstract: Reasoning-acting frameworks enhance large language models (LLMs) by interleaving reasoning with actions for dynamic information acquisition. However, extending this paradigm to graph learning remains underexplored. Graph data...

<https://arxiv.org/abs/2605.07357>

3 CONF

M³: Reframing Training Measures for Discretized Physical Simulations

ArXiv CS.AI 50%

arXiv:2605.08843v1 Announce Type: new Abstract: Neural surrogate models for physical simulations are trained on discretized samples of continuous domains, where the induced empirical measure leads to uneven supervision, biasing optimization and causing spatial inconsistencies in...

<https://arxiv.org/abs/2605.08843>

4 CONF

Ace-Skill: Bootstrapping Multimodal Agents with Prioritized and Clustered Evolution

ArXiv CS.AI 50%

arXiv:2605.08887v1 Announce Type: new Abstract: Self-evolving agents present a promising path toward continual adaptation by distilling task interactions into reusable knowledge artifacts. In practice, this paradigm remains hindered by two coupled bottlenecks: data inefficiency,...

<https://arxiv.org/abs/2605.08887>

5 CONF

Beyond Accuracy: Evaluating Strategy Diversity in LLM Mathematical Reasoning

ArXiv CS.AI 50%

arXiv:2605.09292v1 Announce Type: new Abstract: Large language models now achieve high final-answer accuracy on mathematical reasoning benchmarks, but accuracy alone does not capture reasoning flexibility. We introduce a strategy-level evaluation framework instantiated on 80 AMC...

<https://arxiv.org/abs/2605.09292>

6 CONF

Agentic AI Scientists Are Not Built For Autonomous Scientific Discovery

ArXiv CS.AI 50%

arXiv:2605.08956v1 Announce Type: new Abstract: A growing body of work pursues AI scientists capable of end-to-end autonomous scientific discovery. This position paper argues that although they already function as co-scientists, agentic AI scientists are not built for autonomous...

<https://arxiv.org/abs/2605.08956>

7 CONF

FormalRewardBench: A Benchmark for Formal Theorem Proving Reward Models

ArXiv CS.AI 50%

arXiv:2605.10141v1 Announce Type: new Abstract: Recent neural theorem provers use reinforcement learning with verifiable rewards (RLVR), where proof assistants provide binary correctness signals. While verifiable rewards are cheap and scalable without reward hacking issues, they...

<https://arxiv.org/abs/2605.10141>

8 CONF

SearchSkill: Teaching LLMs to Use Search Tools with Evolving Skill Banks

ArXiv CS.AI 50%

arXiv:2605.09038v1 Announce Type: new Abstract: Teaching language models to use search tools is not only a question of whether they search, but also of whether they issue good queries. This is especially important in open-domain question answering, where broad or copied queries...

<https://arxiv.org/abs/2605.09038>

9 CONF

Combining Mechanical and Agentic Specification Inference for Move

ArXiv CS.AI 50%

arXiv:2605.10005v1 Announce Type: cross Abstract: In this paper, we describe early work on a specification inference tool for the Move Prover that combines a weakest-precondition (WP) analysis over Move bytecode with an agentic coding CLI such as Claude Code. Specification...

<https://arxiv.org/abs/2605.10005>

0 CONF

Token Economics for LLM Agents: A Dual-View Study from Computing and Economics

ArXiv CS.AI 50%

arXiv:2605.09104v1 Announce Type: new Abstract: As LLM agents evolve, tokens have emerged as the core economic primitives of Agentic AI. However, their exponential consumption introduces severe computational, collaborative, and security bottlenecks. Current surveys remain...

<https://arxiv.org/abs/2605.09104>

1 CONF

When Does Non-Uniform Replay Matter in Reinforcement Learning?

ArXiv CS.AI 50%

arXiv:2605.10236v1 Announce Type: cross Abstract: Modern off-policy reinforcement learning algorithms often rely on simple uniform replay sampling and it remains unclear when and why non-uniform replay improves over this strong baseline. Across diverse RL settings, we show that...

<https://arxiv.org/abs/2605.10236>

2 CONF

APCD: Adaptive Path-Contrastive Decoding for Reliable Large Language Model Generation

ArXiv CS.AI 50%

arXiv:2605.09492v1 Announce Type: cross Abstract: Large language models (LLMs) often suffer from hallucinations due to error accumulation in autoregressive decoding, where suboptimal early token choices misguide subsequent generation. Although multi-path decoding can improve...

<https://arxiv.org/abs/2605.09492>

3 CONF

Do LLMs Experience an Internal Polylogue? Investigating Reasoning through the Lens of Personas

ArXiv CS.AI 50%

arXiv:2605.09159v1 Announce Type: new Abstract: Recent work shows that large language models (LLMs) encode behavioural traits ("personas") as linear directions in activation space, often called "persona vectors". Prior work has used such directions as static handles for...

<https://arxiv.org/abs/2605.09159>

4 CONF

Open Ontologies: Tool-Augmented Ontology Engineering with Stable Matching Alignment

ArXiv CS.AI 50%

arXiv:2605.09184v1 Announce Type: new Abstract: We present Open Ontologies, an open-source ontology engineering system implemented in Rust that integrates LLM-driven construction with formal OWL reasoning and ontology alignment via the Model Context Protocol. Our primary finding...

<https://arxiv.org/abs/2605.09184>

5 CONF

Can Agent Benchmarks Support Their Scores? Evidence-Supported Bounds for Interactive-Agent Evaluation

ArXiv CS.AI 50%

arXiv:2605.10448v1 Announce Type: new Abstract: Interactive agent benchmarks map an agent run to a binary outcome through outcome checks. When these checks rely on surface level signals or fail to capture the agent's actual action path, they cannot reliably determine whether the...

<https://arxiv.org/abs/2605.10448>

6 CONF

The Geometry of Forgetting: Temporal Knowledge Drift as an Independent Axis in LLM Representations

ArXiv CS.AI 50%

arXiv:2605.09195v1 Announce Type: new Abstract: Large language models confidently produce outdated answers, and no existing method can detect them. We show this is not an engineering failure but a structural one: temporal drift, whether a stored fact has changed since training,...

<https://arxiv.org/abs/2605.09195>

7 CONF

SeePhys Pro: Diagnosing Modality Transfer and Blind-Training Effects in Multimodal RLVR for Physics Reasoning

ArXiv CS.AI 50%

arXiv:2605.09266v1 Announce Type: new Abstract: We introduce SeePhys Pro, a fine-grained modality transfer benchmark that studies whether models preserve the same reasoning capability when critical information is progressively transferred from text to image. Unlike standard...

<https://arxiv.org/abs/2605.09266>

8 CONF

PoDAR: Power-Disentangled Audio Representation for Generative Modeling

ArXiv CS.AI 50%

arXiv:2605.10084v1 Announce Type: cross Abstract: The performance of audio latent diffusion models is primarily governed by generator expressivity and the modelability of the underlying latent space. While recent research has focused primarily on the former, as well as improving...

<https://arxiv.org/abs/2605.10084>

9 CONF

Towards Conversational Medical AI with Eyes, Ears and a Voice

ArXiv CS.AI 50%

arXiv:2605.09272v1 Announce Type: new Abstract: The practice of medicine relies not only upon skillful dialogue but also on the nuanced exchange and interpretation of rich auditory and visual cues between doctors and patients. Building on the low-latency voice and video...

<https://arxiv.org/abs/2605.09272>

0 CONF

A Prompt-Aware Structuring Framework for Reliable Reuse of AI-Generated Content in the Agentic Web

ArXiv CS.AI 50%

arXiv:2605.09283v1 Announce Type: new Abstract: The evolution of Large Language Models (LLMs) and the software agents built on them (AI agents) marks a turning point in the transition from a human-centric Web to an ``Agentic Web'' driven by AI agents. However, for AI-Generated...

<https://arxiv.org/abs/2605.09283>

1 CONF

How LLMs Are Persuaded: A Few Attention Heads, Rerouted

ArXiv CS.AI 50%

arXiv:2605.09314v1 Announce Type: new Abstract: Language models can be persuaded to abandon factual knowledge. This vulnerability is central to AI safety, but its internal mechanism remains poorly understood. We uncover a compact causal mechanism for persuasion-induced factual...

<https://arxiv.org/abs/2605.09314>

2 CONF

EpiGraph: A Knowledge Graph and Benchmark for Evidence-Intensive Reasoning in Epilepsy

ArXiv CS.AI 50%

arXiv:2605.09505v1 Announce Type: new Abstract: Epilepsy diagnosis and treatment require evidence-intensive reasoning across heterogeneous clinical knowledge, including biosignal patterns, genetic mechanisms, pharmacogenomics, treatment strategies, and patient outcomes. In this...

<https://arxiv.org/abs/2605.09505>

3 CONF

TrajPrism: A Multi-Task Benchmark for Language-Grounded Urban Trajectory Understanding

ArXiv CS.AI 50%

arXiv:2605.10782v1 Announce Type: new Abstract: Urban mobility is naturally expressed both as trajectories in space and as natural-language descriptions of travel intent, constraints, and preferences. However, prior work rarely evaluates these two modalities together on the same...

<https://arxiv.org/abs/2605.10782>

4 CONF

The Wittgensteinian Representation Hypothesis: Is Language the Attractor of Multimodal Convergence?

ArXiv CS.AI 50%

arXiv:2605.09352v1 Announce Type: new Abstract: Understanding why independently trained neural networks from different modalities converge toward shared representations, and where this convergence leads, remains an open question in representation learning. All existing evidence...

<https://arxiv.org/abs/2605.09352>

5 **CONF****Position: Avoid Overstretching LLMs for every Enterprise Task**

ArXiv CS.AI 50%

arXiv:2605.09365v1 Announce Type: new Abstract: Enterprise workloads are dominated by deterministic, structured, and knowledge-dependent tasks operating under strict cost, latency, and reliability constraints. While these are often addressed through large language model (LLM)...

<https://arxiv.org/abs/2605.09365>

6 **CONF****Relations Are Channels: Knowledge Graph Embedding via Kraus Decompositions**

ArXiv CS.AI 50%

arXiv:2605.10317v1 Announce Type: cross Abstract: Knowledge graph embedding (KGE) models typically represent each relation as an operator on entity embeddings. In this work, we identify three structural axioms that any principled relation operator must satisfy, linearity, trace...

<https://arxiv.org/abs/2605.10317>

7 **CONF****PAC-MCTS: Bias-Aware Pruning for Robust LLM-Guided Search and Planning**

ArXiv CS.AI 50%

arXiv:2604.14345v3 Announce Type: replace-cross Abstract: As search depth increases in autonomous reasoning and embodied planning, candidate action spaces expand exponentially, often exhausting computational budgets. While heuristic pruning is a critical countermeasure, existing...

<https://arxiv.org/abs/2604.14345>

8 **CONF****Ister: Linear Transformer for Efficient Multivariate Time Series Forecasting**

ArXiv CS.AI 50%

arXiv:2412.18798v3 Announce Type: replace-cross Abstract: Transformer-based models have achieved remarkable success in multivariate time series forecasting (MTSF) by capturing long-range dependencies. However, their widespread adoption is hindered by the quadratic computational...

<https://arxiv.org/abs/2412.18798>

9 **CONF****Temperature and Persona Shape LLM Agent Consensus With Minimal Accuracy Gains in Qualitative Coding**

ArXiv CS.AI 50%

arXiv:2507.11198v2 Announce Type: replace-cross Abstract: Large Language Models (LLMs) enable new possibilities for qualitative research at scale, including annotation and qualitative coding of educational data. While LLM-based multi-agent systems (MAS) can emulate human coding...

<https://arxiv.org/abs/2507.11198>

10 **CONF****Hypothesis-Driven Deep Research with Large Language Models: A Structured Methodology for Automated Knowledge Discovery**

ArXiv CS.AI 50%

arXiv:2605.10224v1 Announce Type: new Abstract: Current AI-powered research systems adopt a direct search-then-summarize paradigm that treats hypotheses as end products of scientific discovery. We argue this leaves a critical gap: hypotheses can serve a far more powerful role as...

<https://arxiv.org/abs/2605.10224>

1 CONF

PrAg-PO: Prompt Augmented Policy Optimization for Robust and Diverse Mathematical Reasoning

ArXiv CS.AI 50%

arXiv:2602.03190v3 Announce Type: replace-cross Abstract: Reinforcement learning algorithms such as group-relative policy optimization (GRPO) have shown strong potential for improving the mathematical reasoning capabilities of large language models. While a growing body of work...

<https://arxiv.org/abs/2602.03190>

2 CONF

WindINR: Latent-State INR for Fast Local Wind Query and Correction in Complex Terrain

ArXiv CS.AI 50%

arXiv:2605.09511v1 Announce Type: new Abstract: Many downstream decisions in complex terrain require fast wind estimates at a small number of user-specified locations and heights for a given forecast valid time, rather than another dense forecast field on a fixed grid. We...

<https://arxiv.org/abs/2605.09511>

3 CONF

LLM-Guided Monte Carlo Tree Search over Knowledge Graphs: Composing Mechanistic Explanations for Drug-Disease Pairs

ArXiv CS.AI 50%

arXiv:2605.09542v1 Announce Type: new Abstract: Extracting multi-step explanations from knowledge graphs poses a combinatorial challenge requiring both heuristic guidance (as candidates proliferate with depth) and credit assignment (as path quality emerges over extended...

<https://arxiv.org/abs/2605.09542>

4 CONF

FraudBench: A Multimodal Benchmark for Detecting AI-Generated Fraudulent Refund Evidence

ArXiv CS.AI 50%

arXiv:2605.08820v1 Announce Type: cross Abstract: Artificial Intelligence (AI)-generated images have become increasingly realistic and readily adaptable to concrete real-world claims, creating new challenges for verifying visual evidence. A concrete emerging risk is AI-generated...

<https://arxiv.org/abs/2605.08820>

5 CONF

Workspace Optimization: How to Train Your Agent

ArXiv CS.AI 50%

arXiv:2605.09650v1 Announce Type: new Abstract: Modern agents built on frontier language models often cannot adapt their weights. What, then, remains trainable? We argue it is the agent's \emph{workspace}, the structured external substrate it reads, writes, and tests; we call...

<https://arxiv.org/abs/2605.09650>

6 CONF

Absurd World: A Simple Yet Powerful Method to Absurdify the Real-world for Probing LLM Reasoning Capabilities

ArXiv CS.AI 50%

arXiv:2605.09678v1 Announce Type: new Abstract: While extremely powerful and versatile at various tasks, the thinking capabilities of large language models (LLMs) are often put under scrutiny as they sometimes fail to solve problems that humans can systematically solve. However,...

<https://arxiv.org/abs/2605.09678>

7 CONF

Primal-Dual Guided Decoding for Constrained Discrete Diffusion

ArXiv CS.AI 50%

arXiv:2605.09749v1 Announce Type: new Abstract: Discrete diffusion models generate structured sequences by progressively unmasking tokens, but enforcing global property constraints during generation remains an open challenge. We propose primal-dual guided decoding, an...

<https://arxiv.org/abs/2605.09749>

8 CONF

Marrying Generative Model of Healthcare Events with Digital Twin of Social Determinants of Health for Disease Reasoning

ArXiv CS.AI 50%

arXiv:2605.09771v1 Announce Type: new Abstract: Despite the central role of sensor-derived measurements such as imaging traits and plasma biomarkers in biomedical research and clinical practice, existing generative models for disease prediction largely depend on event-level...

<https://arxiv.org/abs/2605.09771>

9 CONF

Energy Consumption of Dataframe Libraries for End-to-End Deep Learning Pipelines:A Comparative Analysis

ArXiv CS.AI 50%

arXiv:2511.08644v3 Announce Type: replace-cross Abstract: This paper presents a detailed comparative analysis of the performance of three major Python data manipulation libraries - Pandas, Polars, and Dask - specifically when embedded within complete deep learning (DL) training...

<https://arxiv.org/abs/2511.08644>

0 CONF

SLASH the Sink: Sharpening Structural Attention Inside LLMs

ArXiv CS.AI 50%

arXiv:2605.10503v1 Announce Type: new Abstract: Large Language Models (LLMs) show remarkable semantic understanding but often struggle with structural understanding when processing graph topologies in a serialized format. Existing solutions rely on training external graph-based...

<https://arxiv.org/abs/2605.10503>

1 CONF

Fairness of Explanations in Artificial Intelligence (AI): A Unifying Framework, Axioms, and Future Direction toward Responsible AI

ArXiv CS.AI 50%

arXiv:2605.09852v1 Announce Type: new Abstract: Machine learning algorithms are being used in high-stakes decisions, including those in criminal justice, healthcare, credit, and employment. The research community has responded with two largely independent research fields:...

<https://arxiv.org/abs/2605.09852>

2 CONF

Probing the Critical Point (CritPt) of AI Reasoning: a Frontier Physics Research Benchmark

ArXiv CS.AI 50%

arXiv:2509.26574v4 Announce Type: replace Abstract: While large language models (LLMs) with reasoning capabilities are progressing rapidly on high-school math competitions and coding, can they reason effectively through complex, open-ended challenges found in frontier physics...

<https://arxiv.org/abs/2509.26574>

3 CONF

Probing Cross-modal Information Hubs in Audio-Visual LLMs

ArXiv CS.AI 50%

arXiv:2605.10815v1 Announce Type: new Abstract: Audio-visual large language models (AVLLMs) have recently emerged as a powerful architecture capable of jointly reasoning over audio, visual, and textual modalities. In AVLLMs, the bidirectional interaction between audio and video...

<https://arxiv.org/abs/2605.10815>

4 CONF

Route by State, Recover from Trace: STAR with Failure-Aware Markov Routing for Multi-Agent Spatiotemporal Reasoning

ArXiv CS.AI 50%

arXiv:2605.10057v1 Announce Type: new Abstract: Compositional spatiotemporal reasoning often requires a system to invoke multiple heterogeneous specialists, such as geometric, temporal, topological, and trajectory agents. A central question is how such a system should route...

<https://arxiv.org/abs/2605.10057>

5 CONF

Continuous Latent Contexts Enable Efficient Online Learning in Transformers

ArXiv CS.AI 50%

arXiv:2605.09867v1 Announce Type: cross Abstract: Large language models (LLMs) exhibit a strong capacity for in-context learning: Given labeled examples, they can generate good predictions without parameter updates. However, many interactive settings go beyond static prediction...

<https://arxiv.org/abs/2605.09867>

6 CONF

Useful for Exploration, Risky for Precision: Evaluating AI Tools in Academic Research

ArXiv CS.AI 50%

arXiv:2605.10125v1 Announce Type: new Abstract: Artificial intelligence (AI) tools are being incorporated into scientific research workflows with the potential to enhance efficiency in tasks such as document analysis, question answering (Q and A), and literature search. However,...

<https://arxiv.org/abs/2605.10125>

7 CONF

Benchmarking Safety Risks of Knowledge-Intensive Reasoning under Malicious Knowledge Editing

ArXiv CS.AI 50%

arXiv:2605.10146v1 Announce Type: new Abstract: Large language models (LLMs) increasingly rely on knowledge editing to support knowledge-intensive reasoning, but this flexibility also introduces critical safety risks: adversaries can inject malicious or misleading knowledge that...

<https://arxiv.org/abs/2605.10146>

8 CONF

TRACE: Distilling Where It Matters via Token-Routed Self On-Policy Alignment

ArXiv CS.AI 50%

arXiv:2605.10194v1 Announce Type: new Abstract: On-policy self-distillation (self-OPD) densifies reinforcement learning with verifiable rewards (RLVR) by letting a policy teach itself under privileged context. We find that when this guidance spans the full response, all-token KL...

<https://arxiv.org/abs/2605.10194>

9 CONF

Towards Autonomous Railway Operations: A Semi-Hierarchical Deep Reinforcement Learning Approach to the Vehicle Rescheduling Problem

ArXiv CS.AI 50%

arXiv:2605.10257v1 Announce Type: new Abstract: Managing disruptions in railway traffic management is a major challenge. Rising traffic density and infrastructure limits increase complexity, making the Vehicle Routing and Scheduling Problem (VRSP) difficult to solve reliably and...

<https://arxiv.org/abs/2605.10257>

00 CONF

IndustryBench: Probing the Industrial Knowledge Boundaries of LLMs

ArXiv CS.AI 50%

arXiv:2605.10267v1 Announce Type: new Abstract: In industrial procurement, an LLM answer is useful only if it survives a standards check: recommended material must match operating condition, every parameter must respect a regulated threshold, and no procedure may contradict a...

<https://arxiv.org/abs/2605.10267>

01 CONF

CORTEG: Foundation Models Enable Cross-Modality Representation Transfer from Scalp to Intracranial Brain Recordings

ArXiv CS.AI 50%

arXiv:2605.10337v1 Announce Type: new Abstract: Intracranial electrocorticography (ECoG) offers high-signal-to-noise access to cortical activity for brain-computer interfaces, yet limited per-patient data has led most prior work to rely on small, subject-specific decoders that...

<https://arxiv.org/abs/2605.10337>

02 CONF

Autonomous FAIR Digital Objects: From Passive Assertions to Active Knowledge

ArXiv CS.AI 50%

arXiv:2605.10370v1 Announce Type: new Abstract: Scientific knowledge on the Web is published as passive assertions and cannot decide when to validate evidence, reconcile contradictions, or update confidence as findings accumulate. Curation depends on centralised middleware and...

<https://arxiv.org/abs/2605.10370>

03 CONF

GraphBench: Next-generation graph learning benchmarking

ArXiv CS.AI 50%

arXiv:2512.04475v5 Announce Type: replace-cross Abstract: Machine learning on graphs has made substantial progress across domains such as molecular property prediction and chip design. Yet benchmarking practices remain fragmented, often relying on narrow, task-specific datasets...

<https://arxiv.org/abs/2512.04475>

04 CONF

GuardAD: Safeguarding Autonomous Driving MLLMs via Markovian Safety Logic

ArXiv CS.AI 50%

arXiv:2605.10386v1 Announce Type: new Abstract: Multimodal large language models (MLLMs) are increasingly integrated into autonomous driving (AD) systems; however, they remain vulnerable to diverse safety threats, particularly in accident-prone scenarios. Recent safeguard...

<https://arxiv.org/abs/2605.10386>

05 CONF

LLM4Branch: Large Language Model for Discovering Efficient Branching Policies of Integer Programs

ArXiv CS.AI 50%

arXiv:2605.10401v1 Announce Type: new Abstract: Efficient branching policies are essential for accelerating Mixed Integer Linear Programming (MILP) solvers. Their design has long relied on hand-crafted heuristics, and now machine learning has emerged as a promising paradigm to...

<https://arxiv.org/abs/2605.10401>

06 CONF

Consistency as a Testable Property: Statistical Methods to Evaluate AI Agent Reliability

ArXiv CS.AI 50%

arXiv:2605.10516v1 Announce Type: new Abstract: This paper establishes a rigorous measurement science for AI agent reliability, providing a foundational framework for quantifying consistency under semantically preserving perturbations. By leveraging χ^2 -statistics for...

<https://arxiv.org/abs/2605.10516>

07 CONF

BEACON: A Multimodal Dataset for Learning Behavioral Fingerprints from Gameplay Data

ArXiv CS.AI 50%

arXiv:2605.10867v1 Announce Type: cross Abstract: Continuous authentication in high-stakes digital environments requires datasets with fine-grained behavioral signals under realistic cognitive and motor demands. But current benchmarks are often limited by small scale, unimodal...

<https://arxiv.org/abs/2605.10867>

08 CONF

A Reflective Storytelling Agent for Older Adults: Integrating Argumentation Schemes and Argument Mining in LLM-Based Personalised Narratives

ArXiv CS.AI 50%

arXiv:2605.10531v1 Announce Type: new Abstract: This work investigates whether knowledge-driven large language model (LLM)-based storytelling can support purposeful narrative interaction with a digital companion for older adults. To address known limitations of LLMs, including...

<https://arxiv.org/abs/2605.10531>

09 CONF

Agent-First Tool API: A Semantic Interface Paradigm for Enterprise AI Agent Systems

ArXiv CS.AI 50%

arXiv:2605.10555v1 Announce Type: new Abstract: As AI agents transition from research prototypes to enterprise production systems, the tool interfaces they consume remain rooted in human-oriented CRUD paradigms. This paper identifies five fundamental architectural mismatches...

<https://arxiv.org/abs/2605.10555>

10 CONF

Beyond the Singular: Revealing the Value of Multiple Generations in Benchmark Evaluation

ArXiv CS.AI 50%

arXiv:2502.08943v4 Announce Type: replace-cross Abstract: Large language models (LLMs) have demonstrated significant utility in real-world applications, exhibiting impressive capabilities in natural language processing and understanding. Benchmark evaluations are crucial for...

<https://arxiv.org/abs/2502.08943>

11 CONF

The Open-Box Fallacy: Why AI Deployment Needs a Calibrated Verification Regime

ArXiv CS.AI 50%

arXiv:2605.10601v1 Announce Type: new Abstract: AI deployment in sensitive domains such as health care, credit, employment, and criminal justice is often treated as unsafe to authorize until model internals can be explained. This often leads to an excessive reliance on...

<https://arxiv.org/abs/2605.10601>

12 CONF

Crystal Fractional Graph Neural Network for Energy Prediction of High-Entropy Alloys

ArXiv CS.AI 50%

arXiv:2605.08103v1 Announce Type: cross Abstract: High-entropy alloys (HEAs) have attracted growing attention for their exceptional mechanical and thermal properties arising from complex atomic configurations. In this paper, we propose crystal fractional graph neural network for...

<https://arxiv.org/abs/2605.08103>

13 CONF

MATRA: Modeling the Attack Surface of Agentic AI Systems -- OpenClaw Case Study

ArXiv CS.AI 50%

arXiv:2605.10763v1 Announce Type: new Abstract: LLMs are increasingly deployed as autonomous agents with access to tools, databases, and external services, yet practitioners (across different sectors) lack systematic methods to assess how known threat classes translate into...

<https://arxiv.org/abs/2605.10763>

14 CONF

Interpretable Machine Learning for Football Performance Analysis: Evidence of Limited Transferability from Elite Leagues to University Competition

ArXiv CS.AI 50%

arXiv:2605.10796v1 Announce Type: new Abstract: Machine learning has become increasingly prevalent in football performance analysis, yet most studies prioritize predictive accuracy while implicitly assuming that learned performance determinants and their interpretations are...

<https://arxiv.org/abs/2605.10796>

15 CONF

New AI-Driven Tools for Enhancing Campus Well-being: A Prevention and Intervention Approach

ArXiv CS.AI 50%

arXiv:2605.10804v1 Announce Type: new Abstract: Campus well-being underpins academic success, yet many universities lack effective methods for monitoring satisfaction and detecting mental health risks. This dissertation addresses these gaps through prevention (improving feedback...

<https://arxiv.org/abs/2605.10804>

16 CONF

Reasoning Is Not Free: Robust Adaptive Cost-Efficient Routing for LLM-as-a-Judge

ArXiv CS.AI 50%

arXiv:2605.10805v1 Announce Type: new Abstract: Reasoning-capable large language models (LLMs) have recently been adopted as automated judges, but their benefits and costs in LLM-as-a-Judge settings remain unclear. Through controlled comparisons between reasoning and...

<https://arxiv.org/abs/2605.10805>

17 CONF

Artificial Intelligence in Number Theory: LLMs for Algorithm Generation and Ensemble Methods for Conjecture Verification

ArXiv CS.AI 50%

arXiv:2504.19451v3 Announce Type: cross Abstract: This paper presents two concrete applications of Artificial Intelligence to algorithmic and analytic number theory. Recent benchmarks of large language models have mainly focused on general mathematics problems and the currently...

<https://arxiv.org/abs/2504.19451>

18 CONF

TTCD:Transformer Integrated Temporal Causal Discovery from Non-Stationary Time Series Data

ArXiv CS.AI 50%

arXiv:2605.08111v1 Announce Type: cross Abstract: The widespread availability of complex time series data in various domains such as environmental science, epidemiology, and economics demands robust causal discovery methods that can identify intricate contemporaneous and lagged...

<https://arxiv.org/abs/2605.08111>

19 CONF

Context-Augmented Code Generation: How Product Context Improves AI Coding Agent Decision Compliance by 49%

ArXiv CS.AI 50%

arXiv:2605.08112v1 Announce Type: cross Abstract: AI coding agents powered by large language models can read codebases and produce functional code, but they routinely violate team-specific product decisions that are invisible in the source code alone. We introduce a controlled...

<https://arxiv.org/abs/2605.08112>

20 CONF

Towards Universal Gene Regulatory Network Inference: Unlocking Generalizable Regulatory Knowledge in Single-cell Foundation Models

ArXiv CS.AI 50%

arXiv:2605.08128v1 Announce Type: cross Abstract: Gene Regulatory Network (GRN) inference is essential for understanding complex cellular mechanisms, rendered tractable through single-cell transcriptomic data. With the emergence of single-cell Foundation Models (scFM), enhanced...

<https://arxiv.org/abs/2605.08128>

21 CONF

Understanding Asynchronous Inference Methods for Vision-Language-Action Models

ArXiv CS.AI 50%

arXiv:2605.08168v1 Announce Type: cross Abstract: Vision-Language-Action (VLA) models offer a promising path to generalist robot control, but their inference latency causes observation staleness when generated actions are executed asynchronously. Several methods have been...

<https://arxiv.org/abs/2605.08168>

22 CONF

Weight Pruning Amplifies Bias: A Multi-Method Study of Compressed LLMs for Edge AI

ArXiv CS.AI 50%

arXiv:2605.08137v1 Announce Type: cross Abstract: Weight pruning is widely advocated for deploying Large Language Models on resource-constrained IoT and edge devices, yet its impact on model fairness remains poorly understood. We conduct a controlled empirical study of three...

<https://arxiv.org/abs/2605.08137>

23 CONF

SlimQwen: Exploring the Pruning and Distillation in Large MoE Model Pre-training

ArXiv CS.AI 50%

arXiv:2605.08738v1 Announce Type: cross Abstract: Structured pruning and knowledge distillation (KD) are typical techniques for compressing large language models, but it remains unclear how they should be applied at pretraining scale, especially to recent mixture-of-experts...

<https://arxiv.org/abs/2605.08738>

24 CONF

Built Environment Reasoning from Remote Sensing Imagery Using Large Vision--Language Models

ArXiv CS.AI 50%

arXiv:2605.08404v1 Announce Type: cross Abstract: This work investigates the use of large language models (LLMs) for tasks in smart cities. The core idea is to leverage remote sensing imagery to characterize the built environment, including design suggestions, constructability...

<https://arxiv.org/abs/2605.08404>

25 CONF

CERSA: Cumulative Energy-Retaining Subspace Adaptation for Memory-Efficient Fine-Tuning

ArXiv CS.AI 50%

arXiv:2605.08174v1 Announce Type: cross Abstract: To mitigate the memory constraints associated with fine-tuning large pre-trained models, existing parameter-efficient fine-tuning (PEFT) methods, such as LoRA, rely on low-rank updates. However, such updates fail to fully capture...

<https://arxiv.org/abs/2605.08174>

26 CONF

LAGO: Language-Guided Adaptive Object-Region Focus for Zero-Shot Visual-Text Alignment

ArXiv CS.AI 50%

arXiv:2605.08156v1 Announce Type: cross Abstract: Zero-shot recognition aims to classify an image by selecting the most compatible label description from a set of candidate classes without any task-specific supervision. In fine-grained settings, however, the relevant evidence...

<https://arxiv.org/abs/2605.08156>

27 CONF

SenseBench: A Benchmark for Remote Sensing Low-Level Visual Perception and Description in Large Vision-Language Models

ArXiv CS.AI 50%

arXiv:2605.10576v1 Announce Type: cross Abstract: Low-level visual perception underpins reliable remote sensing (RS) image analysis, yet current image quality assessment (IQA) methods output uninterpretable scalar scores rather than characterizing physics-driven RS degradations,...

<https://arxiv.org/abs/2605.10576>

28 CONF

When Attention Beats Fourier: Multi-Scale Transformers for PDE Solving on Irregular Domains

ArXiv CS.AI 50%

arXiv:2605.08318v1 Announce Type: cross Abstract: We study the problem of \emph{architecture selection} for deep learning models trained to solve partial differential equations (PDEs), asking when transformer-based architectures with learned attention outperform Fourier-domain...

<https://arxiv.org/abs/2605.08318>

29 CONF

Parameter-Efficient Neuroevolution for Diverse LLM Generation: Quality-Diversity Optimization via Prompt Embedding Evolution

ArXiv CS.AI 50%

arXiv:2605.09781v1 Announce Type: cross Abstract: Large Language Models exhibit mode collapse, producing homogeneous outputs that fail to explore valid solution spaces. We present QD-LLM, a framework for parameter-efficient neuroevolution that evolves prompt embeddings, compact...

<https://arxiv.org/abs/2605.09781>

30 CONF

Information Density as a Quantitative Measure for AI-enabled Virtual Sensing: Feasibility and Limits

ArXiv CS.AI 50%

arXiv:2605.08180v1 Announce Type: cross Abstract: Modern IoT and sensor networks generate vast amounts of data, posing significant challenges for storage, transmission, and real-time processing. Traditional approaches, such as compressive sensing and machine learning-based...

<https://arxiv.org/abs/2605.08180>

31 CONF

Improving TMS EEG Signal Quality for Closed-Loop Neuro Stimulation via Source-Domain Denoising

ArXiv CS.AI 50%

arXiv:2605.08184v1 Announce Type: cross Abstract: This research addresses a validated TMS EEG cleaning pipeline and a corresponding benchmark dataset. It evaluates two widely used artifact removal pipelines. A reference dataset of carefully preprocessed EEG signals was...

<https://arxiv.org/abs/2605.08184>

32 CONF

Neuroscience-Inspired Analyses of Visual Interestingness in Multimodal Transformers

ArXiv CS.AI 50%

arXiv:2605.08188v1 Announce Type: cross Abstract: Human attention is the gateway to conscious perception, memory and decision-making. However, its role in modern transformer models remains largely unexplored. As these systems increasingly influence what people see, prefer and...

<https://arxiv.org/abs/2605.08188>

33 CONF

NeurIPS Should Require Reproducibility Standards for Frontier AI Safety Claims

ArXiv CS.AI 50%

arXiv:2605.08192v1 Announce Type: cross Abstract: Frontier AI safety claims - published assertions that a highly capable general-purpose model is below a threshold of concern, adequately mitigated, or suitable for release - increasingly shape model deployment, governance, and...

<https://arxiv.org/abs/2605.08192>

34 CONF

FairHealth: An Open-Source Python Library for Trustworthy Healthcare AI in Low-Resource Settings

ArXiv CS.AI 50%

arXiv:2605.08198v1 Announce Type: cross Abstract: We present FairHealth, an open-source Python library that provides a unified, modular framework for trustworthy machine learning in healthcare applications, with particular focus on low-resource and low-income country (LMIC)...

<https://arxiv.org/abs/2605.08198>

35 CONF

Beyond Penalization: Diffusion-based Out-of-Distribution Detection and Selective Regularization in Offline Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2605.08202v1 Announce Type: cross Abstract: Offline reinforcement learning (RL) faces a critical challenge of overestimating the value of out-of-distribution (OOD) actions. Existing methods mitigate this issue by penalizing unseen samples, yet they fail to accurately...

<https://arxiv.org/abs/2605.08202>

36 CONF

Bangla-WhisperDiar: Fine-Tuning Whisper and PyAnnote for Bangla Long-Form Speech Recognition and Speaker Diarization

ArXiv CS.AI 50%

arXiv:2605.08214v1 Announce Type: cross Abstract: Automatic Speech Recognition (ASR) and speaker diarization in Bangla remain challenging due to long form recordings, diverse acoustic conditions, and significant speaker variability. This work addresses these two core tasks in...

<https://arxiv.org/abs/2605.08214>

37 CONF

TRAM: Training Approximate Multiplier Structures for Low-Power AI Accelerators

ArXiv CS.AI 50%

arXiv:2605.08231v1 Announce Type: cross Abstract: Reducing power consumption in AI accelerators is increasingly important. Approximate computing can reduce power consumption while keeping the accuracy loss small. Since multipliers are power-hungry components in AI models, this...

<https://arxiv.org/abs/2605.08231>

38 CONF

Explainable Machine Learning Framework for Cardiovascular Disease Diagnosis and Prognosis

ArXiv CS.AI 50%

arXiv:2507.11185v2 Announce Type: replace-cross Abstract: Heart disease continues to pose a critical worldwide health issue, more specifically in areas with insufficient access to healthcare infrastructure and diagnostic systems. Conventional diagnostic approaches often fall...

<https://arxiv.org/abs/2507.11185>

39 CONF

AIPO: : Learning to Reason from Active Interaction

ArXiv CS.AI 50%

arXiv:2605.08401v1 Announce Type: cross Abstract: Recent advances in large language models (LLMs) have demonstrated remarkable reasoning capabilities, largely stimulated by Reinforcement Learning with Verifiable Rewards (RLVR). However, existing RL algorithms face a fundamental...

<https://arxiv.org/abs/2605.08401>

40 CONF

Why Do DiT Editors Drift? Plug-and-Play Low Frequency Alignment in VAE Latent Space

ArXiv CS.AI 50%

arXiv:2605.08250v1 Announce Type: cross Abstract: Recent advances in diffusion transformers (DiTs) have enabled promising single-turn image editing capabilities. However, multi-turn editing often leads to progressive semantic drift and quality degradation. In this work, we study...

<https://arxiv.org/abs/2605.08250>

41 CONF

Can LLMs Predict Polymer Physics Just by Reading Synthesis and Processing Prose?

ArXiv CS.AI 50%

arXiv:2605.08255v1 Announce Type: cross Abstract: Can large language models predict physical and mechanical polymer properties simply by reading unstructured scientific prose? Polymer performance is rarely determined by chemical structure alone; identical nominal polymers can...

<https://arxiv.org/abs/2605.08255>

42 CONF

Research on Security Enhancement Methods for Adversarial Robust Large Language Model Intelligent Agents for Medical Decision-Making Tasks

ArXiv CS.AI 50%

arXiv:2605.08257v1 Announce Type: cross Abstract: Motivated by the challenge to improve the adversarial robustness, security, and trust of medical decision making intelligent agents, this study develops a full-link security enhancement framework, which describes "input risk..."

<https://arxiv.org/abs/2605.08257>

43 CONF

Execution Envelopes: A Shared Admission Contract for Backend AI Execution Requests

ArXiv CS.AI 50%

arXiv:2605.08267v1 Announce Type: cross Abstract: Enterprise AI backends increasingly admit heterogeneous execution requests across model deployment, inference, evaluation, data movement, and agentic workflows. In many systems, those requests arrive in service-specific shapes,...

<https://arxiv.org/abs/2605.08267>

44 CONF

Insider Attacks in Multi-Agent LLM Consensus Systems

ArXiv CS.AI 50%

arXiv:2605.08268v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly deployed in multi-agent systems where agents communicate in natural language to solve tasks jointly. A key capability in such systems is consensus formation, where agents iteratively...

<https://arxiv.org/abs/2605.08268>

45 CONF

SAFormer: Improving Spiking Transformer via Active Predictive Filtering

ArXiv CS.AI 50%

arXiv:2605.08270v1 Announce Type: cross Abstract: Spiking Neural Networks (SNNs) offer notable advantages in biological plausibility and energy efficiency, making them promising candidates for building low-power Transformers. However, existing Spiking Transformers largely adhere...

<https://arxiv.org/abs/2605.08270>

46 CONF

Mitigating Many-shot Jailbreak Attacks with One Single Demonstration

ArXiv CS.AI 50%

arXiv:2605.08277v1 Announce Type: cross Abstract: Many-shot jailbreaking (MSJ) causes safety-aligned language models to answer harmful queries by preceding them with many harmful question-answer demonstrations. We study why this attack becomes stronger as the number of...

<https://arxiv.org/abs/2605.08277>

47 CONF

A Paired Point-of-Care Ultrasound Dataset for Image Quality Enhancement and Benchmarking via a cGAN Baseline

ArXiv CS.AI 50%

arXiv:2605.08282v1 Announce Type: cross Abstract: Purpose: We aim to enhance the image quality of point-of-care ultrasound (POCUS) devices using deep learning and a novel paired dataset of POCUS and high-end ultrasound images. Approach: We collected the first accurately paired...

<https://arxiv.org/abs/2605.08282>

48 CONF

FQPDR: Federated Quantum Neural Network for Privacy-preserving Early Detection of Diabetic Retinopathy

ArXiv CS.AI 50%

arXiv:2605.08324v1 Announce Type: cross Abstract: Diabetic Retinopathy (DR) is a common complication of diabetes that can lead to blindness of people. Detecting DR at the earliest stage is essential to prevent irreversible eye damage. Microaneurysm dots are the first signs of...

<https://arxiv.org/abs/2605.08324>

49 CONF

Graph Computation Meets Circuit Algebra: A Task-Aligned Analysis of Graph Neural Networks for Electronic Design Automation

ArXiv CS.AI 50%

arXiv:2605.08291v1 Announce Type: cross Abstract: EDA problems are graph-structured, but not all graph-structured problems call for the same GNN computation. We argue that successful GNN-for-EDA methods are those whose propagation, aggregation, and supervision align with the...

<https://arxiv.org/abs/2605.08291>

50 CONF

mHC-SSM: Manifold-Constrained Hyper-Connections for State Space Language Models with Stream-Specialized Adapters

ArXiv CS.AI 50%

arXiv:2605.08300v1 Announce Type: cross Abstract: Manifold-Constrained Hyper-Connections (mHC) introduce a stability-motivated variant of multi stream residual mixing by constraining residual stream mixing matrices to the manifold of doubly stochastic matrices via Sinkhorn-Knopp...

<https://arxiv.org/abs/2605.08300>

51 CONF

LLMSYS-HPOBench: Hyperparameter Optimization Benchmark Suite for Real-World LLM Systems

ArXiv CS.AI 50%

arXiv:2605.08305v1 Announce Type: cross Abstract: Large Language Model (LLM) systems have been the frontier of AI in many application domains, leading to new challenges and opportunities for hyperparameter optimization (HPO) for the AutoML community. However, this type of system...

<https://arxiv.org/abs/2605.08305>

52 CONF

Seed Hijacking of LLM Sampling and Quantum Random Number Defense

ArXiv CS.AI 50%

arXiv:2605.08313v1 Announce Type: cross Abstract: Large language models (LLMs) rely on deterministic pseudorandom number generators (PRNGs) for autoregressive sampling, creating a critical supply-chain attack surface overlooked by existing defenses. We present SeedHijack, a...

<https://arxiv.org/abs/2605.08313>

53 CONF

FlashSVD v1.5: Making Low-Rank Transformers Inference Actually Fast

ArXiv CS.AI 50%

arXiv:2605.08314v1 Announce Type: cross Abstract: SVD-based Low-rank compression reduces transformer parameters and nominal FLOPs, but these savings often translate poorly into real LLM serving speedups. We show that this gap is largely a runtime problem: factorized checkpoints...

<https://arxiv.org/abs/2605.08314>

54 CONF

LLM Wardens: Mitigating Adversarial Persuasion with Third-Party Conversational Oversight

ArXiv CS.AI 50%

arXiv:2605.08321v1 Announce Type: cross Abstract: LLMs are increasingly capable of persuasion, which raises the question of how to protect users against manipulation. In a preregistered user study (N=120) across four decision-making scenarios, we find that an adversarial LLM...

<https://arxiv.org/abs/2605.08321>

55 CONF

LLM Advertisement based on Neuron Auctions

ArXiv CS.AI 50%

arXiv:2605.08326v1 Announce Type: cross Abstract: As Large Language Models (LLMs) transition into conversational agents, generative advertising emerges as a crucial monetization strategy. However, embedding advertisements within unstructured LLM outputs introduces a critical...

<https://arxiv.org/abs/2605.08326>

56 CONF

NeuroGAN-3D: Enhancing Intrinsic Functional Brain Networks via High-Fidelity 3D Generative Super-Resolution

ArXiv CS.AI 50%

arXiv:2605.08373v1 Announce Type: cross Abstract: Recent advances in neuroimaging have deepened our understanding of the brain's complex functional and structural organization. Among these, functional Magnetic Resonance Imaging (fMRI) - particularly resting-state fMRI (rs-fMRI)...

<https://arxiv.org/abs/2605.08373>

57 CONF

Reinforcement Learning with Action Chunking

ArXiv CS.AI 50%

arXiv:2507.07969v4 Announce Type: replace-cross Abstract: We present Q-chunking, a simple yet effective recipe for improving reinforcement learning (RL) algorithms for long-horizon, sparse-reward tasks. Our recipe is designed for the offline-to-online RL setting, where the goal...

<https://arxiv.org/abs/2507.07969>

58 CONF

What Software Engineering Looks Like to AI Agents? -- An Empirical Study of AI-Only Technical Discourse on MoltBook

ArXiv CS.AI 50%

arXiv:2605.08380v1 Announce Type: cross Abstract: AI agents are increasingly framed as software-engineering teammates, yet most research studies them inside human-centered workflows. Little is known about the software-engineering discourse autonomous AI agents produce when they...

<https://arxiv.org/abs/2605.08380>

59 **CONF****Effective Explanations Support Planning Under Uncertainty**

ArXiv CS.AI 50%

arXiv:2605.08406v1 Announce Type: cross Abstract: Explaining how to get from A to B can be challenging. It requires mentally simulating what the listener will do based on what they are told. To capture this process, we propose a computational model that converts utterances into...

<https://arxiv.org/abs/2605.08406>

60 **CONF****Fitting Is Not Enough: Smoothness in Extremely Quantized LLMs**

ArXiv CS.AI 50%

arXiv:2605.08894v1 Announce Type: cross Abstract: Large language models (LLMs) achieve strong performance but incur high deployment costs, motivating extremely low-bit but lossy quantization. Existing quantization algorithms mainly focus on improving the numerical accuracy of...

<https://arxiv.org/abs/2605.08894>

61 **CONF****Defense effectiveness across architectural layers: a mechanistic evaluation of persistent memory attacks on stateful LLM agents**

ArXiv CS.AI 50%

arXiv:2605.08442v1 Announce Type: cross Abstract: Persistent memory attacks against LLM agents achieve high attack success rates against open-source models. In these attacks, malicious instructions injected via RAG-retrieved documents are stored in persistent memory and executed...

<https://arxiv.org/abs/2605.08442>

62 **CONF****Do Benchmarks Underestimate LLM Performance? Evaluating Hallucination Detection With LLM-First Human-Adjudicated Assessment**

ArXiv CS.AI 50%

arXiv:2605.08462v1 Announce Type: cross Abstract: Hallucination remains a persistent challenge in Large Language Models (LLMs), particularly in context-grounded settings such as RAG and agentic AI systems. This study focuses on contextual hallucination detection in summarization...

<https://arxiv.org/abs/2605.08462>

63 **CONF****PYTHALAB-MERA: Validation-Grounded Memory, Retrieval, and Acceptance Control for Frozen-LLM Coding Agents**

ArXiv CS.AI 50%

arXiv:2605.08468v1 Announce Type: cross Abstract: Local LLM-based coding agents increasingly work in settings where correctness is earned through execution feedback, persistent state, and bounded repair, not through a single fluent answer. Static retrieval, long-context...

<https://arxiv.org/abs/2605.08468>

64 **CONF****Transformers Can Implement Preconditioned Richardson Iteration for In-Context Gaussian Kernel Regression**

ArXiv CS.AI 50%

arXiv:2605.08475v1 Announce Type: cross Abstract: Mechanistic accounts of in-context learning (ICL) have identified iterative algorithms for linear regression and related linear prediction tasks, often using linear or ReLU attention variants. For nonlinear ICL, prior work has...

<https://arxiv.org/abs/2605.08475>

65 CONF

MathConstraint: Automated Generation of Verified Combinatorial Reasoning Instances for LLMs

ArXiv CS.AI 50%

arXiv:2605.08498v1 Announce Type: cross Abstract: We introduce MathConstraint, a hard, adaptive benchmark for evaluating the combinatorial reasoning capabilities of LLMs. We combine constraint satisfaction problems with rigorous solver-based verification and design an adaptive...

<https://arxiv.org/abs/2605.08498>

66 CONF

Scaling Limits of Long-Context Transformers

ArXiv CS.AI 50%

arXiv:2605.08505v1 Announce Type: cross Abstract: We study the long-context limit of softmax self-attention with a fixed query and a random context of n i.i.d. keys on the sphere, viewing the inverse temperature β_n as the scaling parameter that decides whether attention...

<https://arxiv.org/abs/2605.08505>

67 CONF

RW-Post: Auditable Evidence-Grounded Multimodal Fact-Checking in the Wild

ArXiv CS.AI 50%

arXiv:2605.10357v1 Announce Type: cross Abstract: Multimodal misinformation increasingly leverages visual persuasion, where repurposed or manipulated images strengthen misleading text. We introduce RW-Post , a post-aligned text-image benchmark for real-world...

<https://arxiv.org/abs/2605.10357>

68 CONF

Geometry Guided Self-Consistency for Physical AI

ArXiv CS.AI 50%

arXiv:2605.08638v1 Announce Type: cross Abstract: State-of-the-art physical AI models generate a chunk of actions per inference through diffusion or flow matching, iteratively refining an initial noise sample into an action trajectory. Because this inference process is...

<https://arxiv.org/abs/2605.08638>

69 CONF

Sketch-and-Verify: Structured Inference-Time Scaling via Program Sketching

ArXiv CS.AI 50%

arXiv:2605.08658v1 Announce Type: cross Abstract: SKETCHVERIFY is a within-tier cost-performance policy, not a universal accuracy improvement. The operational question: a practitioner stuck with a small, cheap code model (here, Gemini 3.1 Flash Lite) for latency, deployment, or...

<https://arxiv.org/abs/2605.08658>

70 CONF

Explanation Fairness in Large Language Models: An Empirical Analysis of Disparities in How LLMs Justify Decisions Across Demographic Groups

ArXiv CS.AI 50%

arXiv:2605.08671v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly deployed not only to make decisions but to explain them. While AI decision fairness has been studied extensively, the fairness of AI explanations (whether LLMs justify decisions with...

<https://arxiv.org/abs/2605.08671>

71 CONF

ChaosNetBench: Benchmarking Spatio-Temporal Graph Neural Networks on Chaotic Lattice Dynamics

ArXiv CS.AI 50%

arXiv:2605.09676v1 Announce Type: cross Abstract: Spatio-temporal graph neural networks (STGNNs) are widely used for short-term forecasting in dynamic physical systems such as traffic and weather. However, the prevailing evaluation practice uses real world benchmark data sets in...

<https://arxiv.org/abs/2605.09676>

72 CONF

Structure-Centric Graph Foundation Model via Geometric Bases

ArXiv CS.AI 50%

arXiv:2605.08689v1 Announce Type: cross Abstract: Graph foundation models (GFMs) seek transferable representations across graph domains but are limited by structural heterogeneity and incompatible node feature spaces. We propose Structure-Centric Graph Foundation Models (SCGFM),...

<https://arxiv.org/abs/2605.08689>

73 CONF

ALAM: Algebraically Consistent Latent Transitions for Vision-Language-Action Models

ArXiv CS.AI 50%

arXiv:2605.10819v1 Announce Type: cross Abstract: Vision-language-action (VLA) models remain constrained by the scarcity of action-labeled robot data, whereas action-free videos provide abundant evidence of how the physical world changes. Latent action models offer a promising...

<https://arxiv.org/abs/2605.10819>

74 CONF

Debugging the Debuggers: Failure-Anchored Structured Recovery for Software Engineering Agents

ArXiv CS.AI 50%

arXiv:2605.08717v1 Announce Type: cross Abstract: Software engineering agents are increasingly deployed in evaluable engineering environments, yet post-failure recovery remains costly, manual, and ad hoc. Existing systems expose traces or generate follow-up feedback, but they do...

<https://arxiv.org/abs/2605.08717>

75 CONF

Omni-scale Learning-based Sequential Decision Framework for Order Fulfillment of Tote-handling Robotic Systems

ArXiv CS.AI 50%

arXiv:2605.08758v1 Announce Type: cross Abstract: Driven by the rapid expansion of e-commerce and small-batch production, the size of the intralogistics load unit of finished goods, semi-finished goods and raw materials is steadily shrinking. Totes are gradually replacing...

<https://arxiv.org/abs/2605.08758>

76 CONF

Deterministic Decomposition of Stochastic Generative Dynamics

ArXiv CS.AI 50%

arXiv:2605.08794v1 Announce Type: cross Abstract: Modern generative models can be understood as probability transport from a simple base distribution to a target data distribution. Deterministic transport models offer tractable velocity-field parameterizations, whereas...

<https://arxiv.org/abs/2605.08794>

77 CONF

Curvature-Aware Captioning:Leveraging Geodesic Attention for 3D Scene Understanding

ArXiv CS.AI 50%

arXiv:2605.08808v1 Announce Type: cross Abstract: Accurate 3D scene description is fundamental to robotic navigation and augmented reality, yet current dense captioning methods face significant limitations in processing sparse point cloud data. % Existing approaches that apply...

<https://arxiv.org/abs/2605.08808>

78 CONF

SimReg: Achieving Higher Performance in the Pretraining via Embedding Similarity Regularization

ArXiv CS.AI 50%

arXiv:2605.08809v1 Announce Type: cross Abstract: Pretraining large language models (LLMs) with next-token prediction has led to remarkable advances, yet the context-dependent nature of token embeddings in such models results in high intra-class variance and inter-class...

<https://arxiv.org/abs/2605.08809>

79 CONF

The Grounding Gap: How LLMs Anchor the Meaning of Abstract Concepts Differently from Humans

ArXiv CS.AI 50%

arXiv:2605.08837v1 Announce Type: cross Abstract: Abstract concepts - justice, theory, availability - have no single perceivable referent; in the human brain, their meaning emerges from a web of experiences, affect, and social context. Do large language models (LLMs) ground...

<https://arxiv.org/abs/2605.08837>

80 CONF

Generating Leakage-Free Benchmarks for Robust RAG Evaluation

ArXiv CS.AI 50%

arXiv:2605.08838v1 Announce Type: cross Abstract: Retrieval-augmented generation (RAG) is widely used to augment large language models (LLMs) with external knowledge. However, many benchmark datasets, designed to test RAG performance, comprise many questions that can already be...

<https://arxiv.org/abs/2605.08838>

81 CONF

Geometrically Constrained Stenosis Editing in Coronary Angiography via Entropic Optimal Transport

ArXiv CS.AI 50%

arXiv:2605.08851v1 Announce Type: cross Abstract: The scarcity of high-quality imaging data for coronary angiography (CAG) stenosis limits the clinical translation of automated stenosis detection. Synthetic stenosis data provides a practical avenue to augment training sets,...

<https://arxiv.org/abs/2605.08851>

82 CONF

BubbleSpec: Turning Long-Tail Bubbles into Speculative Rollout Drafts for Synchronous Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2605.08862v1 Announce Type: cross Abstract: Reinforcement Learning (RL) has become a cornerstone for improving the performance of Large Language Models (LLMs). However, its rollout phase constitutes a significant efficiency bottleneck, mainly arising from the long-tail...

<https://arxiv.org/abs/2605.08862>

83 CONF

Why Do Aligned LLMs Remain Jailbreakable: Refusal-Escape Directions, Operator-Level Sources, and Safety-Utility Trade-off

ArXiv CS.AI 50%

arXiv:2605.08878v1 Announce Type: cross Abstract: Aligned large language models (LLMs) remain vulnerable to jailbreak attacks. Recent mechanistic studies have identified latent features and representation shifts associated with jailbreak success, but they leave a more...

<https://arxiv.org/abs/2605.08878>

84 CONF

FragileFlow: Spectral Control of Correct-but-Fragile Predictions for Foundation Model Robustness

ArXiv CS.AI 50%

arXiv:2605.08896v1 Announce Type: cross Abstract: Robust adaptation of LLMs and VLMs is often evaluated by average accuracy or average consistency under perturbations. However, these averages can hide a structured failure mode: a prediction may remain correct while probability...

<https://arxiv.org/abs/2605.08896>

85 CONF

DAPE: Dynamic Non-uniform Alignment and Progressive Detail Enhancement Techniques for Improving the Performance of Efficient Visual Language Models

ArXiv CS.AI 50%

arXiv:2605.08902v1 Announce Type: cross Abstract: In recent years, pre-trained visual-linguistic models have demonstrated tremendous potential, becoming a crucial foundational framework for numerous downstream tasks. However, the information density between text and images is...

<https://arxiv.org/abs/2605.08902>

86 CONF

Transformer autoencoder with local attention for sparse and irregular time series with application on risk estimation

ArXiv CS.AI 50%

arXiv:2605.08914v1 Announce Type: cross Abstract: This paper introduces a framework specifically designed for sparse and irregular time series {risk estimation}. It is based on a Transformer Autoencoder with local attention, which leverages the powerful pattern identification...

<https://arxiv.org/abs/2605.08914>

87 CONF

A Quantum Inspired Variational Kernel and Explainable AI Framework for Cross Region Solar and Wind Energy Forecasting

ArXiv CS.AI 50%

arXiv:2605.09032v1 Announce Type: cross Abstract: Reliable short horizon forecasting of solar and wind generation is a structural prerequisite of any modern power system yet most published forecasters are tuned and evaluated on a single climatic regime and most algorithmic...

<https://arxiv.org/abs/2605.09032>

88 CONF

Octopus Protocol: One-Shot Hardware Discovery and Control for AI Agents via Infrastructure-as-Prompts

ArXiv CS.AI 50%

arXiv:2605.09055v1 Announce Type: cross Abstract: Recent agentic-robotics systems, from Code-asPolicies to modern vision-language-action (VLA) foundation models, presuppose that drivers, SDKs, or ROS-style primitives for the target hardware already exist. Writing those...

<https://arxiv.org/abs/2605.09055>

89 CONF

Single-Configuration Attack Success Rate Is Not Enough: Jailbreak Evaluations Should Report Distributional Attack Success

ArXiv CS.AI 50%

arXiv:2605.09070v1 Announce Type: cross Abstract: Many jailbreak attack research papers report attack success rates for a limited number of parameter settings, even though there are many combinations of parameter settings that could be used. Further, when new jailbreak papers...

<https://arxiv.org/abs/2605.09070>

90 CONF

Robust Multi-Agent LLMs under Byzantine Faults

ArXiv CS.AI 50%

arXiv:2605.09076v1 Announce Type: cross Abstract: Large language model (LLM) agents increasingly collaborate over peer-to-peer networks to improve their reliability. However, these same interactions can also become a source of vulnerability, as unreliable or Byzantine agents may...

<https://arxiv.org/abs/2605.09076>

91 CONF

Biosignal Fingerprinting: A Cross-Modal PPG-ECG Foundation Model

ArXiv CS.AI 50%

arXiv:2605.09579v1 Announce Type: cross Abstract: Cardiovascular disease remains the leading cause of global mortality, yet scalable cardiac monitoring is hindered by the gap between diagnostic-rich ECG and ubiquitous wearable PPG. Bridging this gap requires representations that...

<https://arxiv.org/abs/2605.09579>

92 CONF

AI Native Asset Intelligence

ArXiv CS.AI 50%

arXiv:2605.09115v1 Announce Type: cross Abstract: Modern security environments generate fragmented signals across cloud resources, identities, configurations, and third-party security tools. Although AI-native security assistants improve access to this data, they remain largely...

<https://arxiv.org/abs/2605.09115>

93 CONF

A Versatile AI Agent for Rare Disease Diagnosis and Risk Gene Prioritization

ArXiv CS.AI 50%

arXiv:2605.06226v2 Announce Type: replace Abstract: Accurate and timely diagnosis is essential for effective treatment, particularly in the context of rare diseases. However, current diagnostic workflows often lead to prolonged assessment times and low accuracy. To address these...

<https://arxiv.org/abs/2605.06226>

94 CONF

From Traditional Taggers to LLMs: A Comparative Study of POS Tagging for Medieval Romance Languages

ArXiv CS.AI 50%

arXiv:2605.09147v1 Announce Type: cross Abstract: Part-of-speech (POS) tagging for Medieval Romance languages remains challenging due to orthographic variation, morphological complexity, and limited annotated resources. This paper presents a systematic empirical evaluation of...

<https://arxiv.org/abs/2605.09147>

95 CONF

Navigating LLM Valley: From AdamW to Memory-Efficient and Matrix-Based Optimizers

ArXiv CS.AI 50%

arXiv:2605.09176v1 Announce Type: cross Abstract: Training large language models requires optimization algorithms that are not only statistically effective, but also computationally and memory efficient at extreme scale. Although Adam remains the dominant optimizer for...

<https://arxiv.org/abs/2605.09176>

96 CONF

DARE: Difficulty-Adaptive Reinforcement Learning with Co-Evolved Difficulty Estimation

ArXiv CS.AI 50%

arXiv:2605.09188v1 Announce Type: cross Abstract: Reinforcement learning improves the reasoning ability of large language models but remains costly and sample-inefficient, as many rollouts provide weak learning signals. Difficulty-aware data selection methods attempt to address...

<https://arxiv.org/abs/2605.09188>

97 CONF

Detect, Localize, and Explain: Interactive Hierarchical Log Anomaly Analytics with LLM Augmentation

ArXiv CS.AI 50%

arXiv:2605.09222v1 Announce Type: cross Abstract: Logs are ubiquitous in modern systems. Unfortunately, their unstructured nature in flat sequences limits understanding of execution behaviors, hindering effective anomaly diagnosis. To address this, Krone introduces a novel...

<https://arxiv.org/abs/2605.09222>

98 CONF

The Art of the Jailbreak: Formulating Jailbreak Attacks for LLM Security Beyond Binary Scoring

ArXiv CS.AI 50%

arXiv:2605.09225v1 Announce Type: cross Abstract: Jailbreak attacks -- adversarial prompts that bypass LLM alignment through purely linguistic manipulation -- pose a growing operational security threat, yet the field lacks large-scale, reproducible infrastructure for generating,...

<https://arxiv.org/abs/2605.09225>

99 CONF

Monocular Biomechanical Tracking of Fingers with Inverse Kinematics to Foundation Models

ArXiv CS.AI 50%

arXiv:2605.09258v1 Announce Type: cross Abstract: Accurate hand and finger tracking from video has significant clinical applications for monitoring activities of daily living and measuring range of motion, yet monocular video approaches for obtaining hand biomechanics remain...

<https://arxiv.org/abs/2605.09258>

00 CONF

Any2Any 3D Diffusion Models with Knowledge Transfer: A Radiotherapy Planning Study

ArXiv CS.AI 50%

arXiv:2605.09622v1 Announce Type: cross Abstract: Voxel-wise dose prediction is a critical yet challenging task in practical radiotherapy (RT) planning, as bespoke models trained from scratch often struggle to generalize across diverse clinical settings. Meanwhile, generative...

<https://arxiv.org/abs/2605.09622>

01 CONF

Beyond Continuity: Challenges of Context Switching in Multi-Turn Dialogue with LLMs

ArXiv CS.AI 50%

arXiv:2605.09268v1 Announce Type: cross Abstract: Users interacting with Large Language Models (LLMs) in a multi-turn conversation routinely refine their requests or pivot to new topics. LLMs, however, often miss these topic shifts and carry over irrelevant context from previous...

<https://arxiv.org/abs/2605.09268>

02 CONF

Gender Fairness in Audio Deepfake Detection: Performance and Disparity Analysis

ArXiv CS.AI 50%

arXiv:2603.09007v2 Announce Type: replace-cross Abstract: Audio deepfake detection aims to detect real human voices from those generated by Artificial Intelligence (AI) and has emerged as a significant problem in the field of voice biometrics systems. With the ever-improving...

<https://arxiv.org/abs/2603.09007>

03 CONF

Turning Drift into Constraint: Robust Reasoning Alignment in Non-Stationary Multi-Stream Environments

ArXiv CS.AI 50%

arXiv:2510.04142v3 Announce Type: replace-cross Abstract: This paper identifies a critical yet underexplored challenge in reasoning alignment from multiple multi-modal large language models (MLLMs): In non-stationary environments, the diverse reasoning distributions of source...

<https://arxiv.org/abs/2510.04142>

04 CONF

Your Simulation Runs but Solves the Wrong Physics: PDE-Grounded Intent Verification for LLM-Generated Multiphysics Simulation Code

ArXiv CS.AI 50%

arXiv:2605.09360v1 Announce Type: cross Abstract: Execution-based evaluation of LLM-generated code implicitly treats successful execution as a proxy for correctness. In scientific simulation, this proxy is insufficient: a generated input file can run, mesh, and converge while...

<https://arxiv.org/abs/2605.09360>

05 CONF

From Detection to Recovery: Operational Analysis on LLM Pre-training with 504 GPUs

ArXiv CS.AI 50%

arXiv:2605.09370v1 Announce Type: cross Abstract: Large-scale AI training is now fundamentally a distributed systems problem, and hardware failures have become routine operating conditions rather than rare exceptions. Public operational evidence from production training...

<https://arxiv.org/abs/2605.09370>

06 CONF

AtteConDA: Attention-Based Conflict Suppression in Multi-Condition Diffusion Models and Synthetic Data Augmentation

ArXiv CS.AI 50%

arXiv:2605.09425v1 Announce Type: cross Abstract: Recent conditional image generation methods can improve controllability by generating images that are faithful to conditions such as sketches, human poses, segmentation maps, and depth. By applying these techniques to image...

<https://arxiv.org/abs/2605.09425>

07 CONF

Key Coverage Matters: Semi-Structured Extraction of OCR Clinical Reports

ArXiv CS.AI 50%

arXiv:2605.09440v1 Announce Type: cross Abstract: Clinical reports are often fragmented across healthcare institutions because privacy regulations and data silos limit direct information sharing. When patients seek care at a different hospital, they often carry paper or scanned...

<https://arxiv.org/abs/2605.09440>

08 CONF

Outlier-Robust Diffusion Solvers for Inverse Problems

ArXiv CS.AI 50%

arXiv:2605.09477v1 Announce Type: cross Abstract: Methods based on diffusion models (DMs) for solving inverse problems (IPs) have recently achieved remarkable performance. However, DM-based methods typically struggle against outliers, which are common in real-world measurements....

<https://arxiv.org/abs/2605.09477>

09 CONF

Spectral Transformer Neural Processes

ArXiv CS.AI 50%

arXiv:2605.09498v1 Announce Type: cross Abstract: Time series, spatial data, and images are natural applications of Neural Processes. However, when such data exhibit strong periodicity and quasi-periodicity, existing methods often suffer from underfitting and generalise poorly...

<https://arxiv.org/abs/2605.09498>

10 CONF

Hidden Error Awareness in Chain-of-Thought Reasoning: The Signal Is Diagnostic, Not Causal

ArXiv CS.AI 50%

arXiv:2605.09502v1 Announce Type: cross Abstract: Chain-of-thought (CoT) prompting assumes that generated reasoning reflects a model's internal computation. We show this assumption is wrong in a specific, measurable way: models internally detect their own reasoning errors but...

<https://arxiv.org/abs/2605.09502>

11 CONF

Position: AI Security Policy Should Target Systems, Not Models

ArXiv CS.AI 50%

arXiv:2605.09504v1 Announce Type: cross Abstract: We present swarm-attack, an open-source adversarial testing framework in which multiple lightweight LLM agents coordinate through shared memory, parallel exploration, and evolutionary optimization. Together, our results...

<https://arxiv.org/abs/2605.09504>

12 CONF

Assessment of RAG and Fine-Tuning for Industrial Question-Answering-Applications

ArXiv CS.AI 50%

arXiv:2605.09533v1 Announce Type: cross Abstract: Large Language Models (LLMs) are increasingly employed in enterprise question-answering (QA) systems, requiring adaptation to domain-specific knowledge. Among the most prevalent methods for incorporating such knowledge are...

<https://arxiv.org/abs/2605.09533>

13 CONF

Governing AI-Assisted Security Operations: A Design Science Framework for Operational Decision Support

ArXiv CS.AI 50%

arXiv:2605.09534v1 Announce Type: cross Abstract: Engineering managers increasingly must decide how to introduce generative artificial intelligence (AI), retrieval-augmented generation, and coding agents into high-risk operational functions without weakening accountability,...

<https://arxiv.org/abs/2605.09534>

14 CONF

Efficient Ensemble Selection from Binary and Pairwise Feedback

ArXiv CS.AI 50%

arXiv:2605.09588v1 Announce Type: cross Abstract: Organizations increasingly deploy multiple AI systems across task domains, but selecting a small, high-performing ensemble can require costly model calls, benchmark runs, and human evaluation. We study this selection problem as a...

<https://arxiv.org/abs/2605.09588>

15 CONF

Do multimodal models imagine electric sheep?

ArXiv CS.AI 50%

arXiv:2605.09693v1 Announce Type: cross Abstract: Yes. We find that large multimodal models develop mental imagery when solving spatial puzzles, and they do imagine sheep when solving sheep puzzles. We fine-tune a Qwen3.5 VLM to solve twelve diverse visual reasoning tasks --...

<https://arxiv.org/abs/2605.09693>

16 CONF

Metal-Sci: A Scientific Compute Benchmark for Evolutionary LLM Kernel Search on Apple Silicon

ArXiv CS.AI 50%

arXiv:2605.09708v1 Announce Type: cross Abstract: We present Metal-Sci, a 10-task benchmark of scientific Apple Silicon Metal compute kernels spanning six optimization regimes (stencils, all-pairs in n^2 -body problems, multi-field Boltzmann, neighbor-list molecular dynamics,...

<https://arxiv.org/abs/2605.09708>

17 CONF

Distilling 3D Spatial Reasoning into a Lightweight Vision-Language Model with CoT

ArXiv CS.AI 50%

arXiv:2605.09719v1 Announce Type: cross Abstract: Large-scale 3D vision-language models (VLMs) like LLaVA-3D offer strong spatial reasoning but are difficult to deploy due to high computational costs. We propose a knowledge distillation framework that transfers spatial reasoning...

<https://arxiv.org/abs/2605.09719>

18 CONF

Security Risks in Tool-Enabled AI Agents: A Systematic Analysis of Privileged Execution Environments

ArXiv CS.AI 50%

arXiv:2605.09721v1 Announce Type: cross Abstract: Tool-enabled AI agents are increasingly deployed in cloud-hosted environments and offered as services, where they perform side-effecting operations through privileged tools within execution environments. While such agents enable...

<https://arxiv.org/abs/2605.09721>

19 CONF

KV-RM: Regularizing KV-Cache Movement for Static-Graph LLM Serving

ArXiv CS.AI 50%

arXiv:2605.09735v1 Announce Type: cross Abstract: Static-graph LLM decoders provide predictable launches, fixed tensor shapes, and low submission overhead, but online decoding exposes highly irregular KV-cache behavior: request lengths differ, EOS events arrive asynchronously,...

<https://arxiv.org/abs/2605.09735>

20 CONF

The Silent Vote: Improving Zero-Shot LLM Reliability by Aggregating Semantic Neighborhoods

ArXiv CS.AI 50%

arXiv:2605.09739v1 Announce Type: cross Abstract: Large Language Models are increasingly used as zero-shot classifiers in complex reasoning tasks. However, standard constrained decoding suffers from a phenomenon we define as Renormalization Bias. When a model is restricted to a...

<https://arxiv.org/abs/2605.09739>

21 CONF

Exploitation Without Deception: Dark Triad Feature Steering Reveals Separable Antisocial Circuits in Language Models

ArXiv CS.AI 50%

arXiv:2605.09773v1 Announce Type: cross Abstract: We use sparse autoencoder (SAE) feature steering to amplify Dark Triad personality traits (Machiavellianism, narcissism, and psychopathy) in Llama-3.3-70B-Instruct and evaluate the resulting behavioral changes across five...

<https://arxiv.org/abs/2605.09773>

22 CONF

EvoPref: Multi-Objective Evolutionary Optimization Discovers Diverse LLM Alignments Beyond Gradient Descent

ArXiv CS.AI 50%

arXiv:2605.09777v1 Announce Type: cross Abstract: Gradient-based preference optimization methods for large language model (LLM) alignment suffer from preference collapse, converging to narrow behavioral modes while neglecting preference diversity. We introduce EvoPref, a...

<https://arxiv.org/abs/2605.09777>

23 CONF

CrossVL: Complexity-Aware Feature Routing and Paired Curriculum for Cross-View Vision-Language Detection

ArXiv CS.AI 50%

arXiv:2605.09802v1 Announce Type: cross Abstract: Vision-language models (VLMs) enable text-guided object detection but degrade severely under cross-view scenarios where ground and aerial viewpoints differ in altitude, scale, and spatial layout. These geometric changes introduce...

<https://arxiv.org/abs/2605.09802>

24 CONF

Insight: Enhancing Mobile Accessibility for Blind and Visually Impaired Users with LLMs

ArXiv CS.AI 50%

arXiv:2605.09803v1 Announce Type: cross Abstract: This research paper addresses the limitations of current mobile accessibility services like TalkBack, which provide manual gesture-based sequential feedback to BVI users. Motivated by the promise of large language models (LLMs),...

<https://arxiv.org/abs/2605.09803>

25 CONF

Fashion Florence: Fine-Tuning Florence-2 for Structured Fashion Attribute Extraction

ArXiv CS.AI 50%

arXiv:2605.09827v1 Announce Type: cross Abstract: We present Fashion Florence, a Florence-2 vision-language model fine-tuned with LoRA to extract structured fashion attributes from clothing images. Given a single photograph, the model generates a JSON object containing category,...

<https://arxiv.org/abs/2605.09827>

26 CONF

Free Energy Manifold: Score-Based Inference for Hybrid Bayesian Networks

ArXiv CS.AI 50%

arXiv:2605.09839v1 Announce Type: cross Abstract: We introduce the Free Energy Manifold (FEM), a score-trained conditional energy model specialized for inference in hybrid Bayesian networks with discrete and continuous variables. FEM represents each conditional factor as an...

<https://arxiv.org/abs/2605.09839>

27 CONF

EgoMemReason: A Memory-Driven Reasoning Benchmark for Long-Horizon Egocentric Video Understanding

ArXiv CS.AI 50%

arXiv:2605.09874v1 Announce Type: cross Abstract: Next-generation visual assistants, such as smart glasses, embodied agents, and always-on life-logging systems, must reason over an entire day or more of continuous visual experience. In ultra-long video settings, relevant...

<https://arxiv.org/abs/2605.09874>

28 CONF

Verifier-Free RL for LLMs via Intrinsic Gradient-Norm Reward

ArXiv CS.AI 50%

arXiv:2605.09920v1 Announce Type: cross Abstract: While Reinforcement Learning with Verifiable Rewards (RLVR) has recently emerged as a promising post-training paradigm for Large Language Models (LLMs), its dependency on the gold label or domain-specific verifiers limits its...

<https://arxiv.org/abs/2605.09920>

29 CONF

Team-Based Self-Play With Dual Adaptive Weighting for Fine-Tuning LLMs

ArXiv CS.AI 50%

arXiv:2605.09922v1 Announce Type: cross Abstract: While recent self-training approaches have reduced reliance on human-labeled data for aligning LLMs, they still face critical limitations: (i) sensitivity to synthetic data quality, leading to instability and bias amplification...

<https://arxiv.org/abs/2605.09922>

30 CONF

PruneTIR: Inference-Time Tool Call Pruning for Effective yet Efficient Tool-Integrated Reasoning

ArXiv CS.AI 50%

arXiv:2605.09931v1 Announce Type: cross Abstract: Tool-integrated reasoning (TIR) enables large language models (LLMs) to enhance their capabilities by interacting with external tools, such as code interpreters (CI). Most recent studies focus on exploring various methods to...

<https://arxiv.org/abs/2605.09931>

31 CONF

HapticLDM: A Diffusion Model for Text-to-Vibrotactile Generation

ArXiv CS.AI 50%

arXiv:2605.09971v1 Announce Type: cross Abstract: Text-to-vibration generation converts natural language into haptic feedback, enabling vibration-effect designers to get scenarios-fitted vibrations more efficiently, which shows great potentials in application fields such as...

<https://arxiv.org/abs/2605.09971>

32 CONF

The two clocks and the innovation window: When and how generative models learn rules

ArXiv CS.AI 50%

arXiv:2605.10019v1 Announce Type: cross Abstract: Generative models trained on finite data face a fundamental tension: their score-matching or next-token objective converges to the empirical training distribution rather than the population distribution we seek to learn. Using...

<https://arxiv.org/abs/2605.10019>

33 CONF

Speech-based Psychological Crisis Assessment using LLMs

ArXiv CS.AI 50%

arXiv:2605.10027v1 Announce Type: cross Abstract: Psychological support hotlines provide critical support for individuals experiencing mental health emergencies, yet current assessments largely rely on human operators whose judgments may vary with professional experience and are...

<https://arxiv.org/abs/2605.10027>

34 CONF

Bridging the Cognitive Gap: A Unified Memory Paradigm for 6G Agentic AI-RAN

ArXiv CS.AI 50%

arXiv:2605.10036v1 Announce Type: cross Abstract: As 6G evolves, the radio access network must transcend traditional automation to embrace agentic AI capable of perception, reasoning, and evolution. A fundamental cognitive gap persists in current disaggregated architectures,...

<https://arxiv.org/abs/2605.10036>

35 CONF

Personalizing LLMs with Binary Feedback: A Preference-Corrected Optimization Framework

ArXiv CS.AI 50%

arXiv:2605.10043v1 Announce Type: cross Abstract: Large Language Model (LLM) personalization aims to align model behaviors with individual user preferences. Existing methods often focus on isolated user histories, neglecting the essential role of inter-user differences. We...

<https://arxiv.org/abs/2605.10043>

36 CONF

Not-So-Strange Love: Language Models and Generative Linguistic Theories are More Compatible than They Appear

ArXiv CS.AI 50%

arXiv:2605.10061v1 Announce Type: cross Abstract: Futrell and Mahowald (2025) frame the success of neural language models (LMs) as supporting gradient, usage-based linguistic theories. I argue that LMs can also instantiate theories based on formal structures - the types of...

<https://arxiv.org/abs/2605.10061>

37 CONF

MicroWorld: Empowering Multimodal Large Language Models to Bridge the Microscopic Domain Gap with Multimodal Attribute Graph

ArXiv CS.AI 50%

arXiv:2605.10120v1 Announce Type: cross Abstract: Multimodal large language models (MLLMs) show remarkable potential for scientific reasoning, yet their performance in specialized domains such as microscopy remains limited by the scarcity of domain-specific training data and the...

<https://arxiv.org/abs/2605.10120>

38 CONF

Explainability of Recurrent Neural Networks for Enhancing P300-based Brain-Computer Interfaces

ArXiv CS.AI 50%

arXiv:2605.10121v1 Announce Type: cross Abstract: Brain-Computer Interfaces (BCIs) based on P300 event-related potentials offer promising applications in health, education, and assistive technologies. However, challenges related to inter- and intra-subject variability and the...

<https://arxiv.org/abs/2605.10121>

39 CONF

When Prompts Become Payloads: A Framework for Mitigating SQL Injection Attacks in Large Language Model-Driven Applications

ArXiv CS.AI 50%

arXiv:2605.10176v1 Announce Type: cross Abstract: Natural language interfaces to structured databases are becoming increasingly common, largely due to advances in large language models (LLMs) that enable users to query data using conversational input rather than formal query...

<https://arxiv.org/abs/2605.10176>

40 CONF

Developing a foundation model for high-resolution remote sensing data of the Netherlands

ArXiv CS.AI 50%

arXiv:2605.10184v1 Announce Type: cross Abstract: We develop a foundation model using 1.2m high resolution satellite images of the Netherlands. By combining a Convolutional Neural Network and a Vision Transformer, the model captures both low- and high-frequency landscape...

<https://arxiv.org/abs/2605.10184>

41 CONF

DynGhost: Temporally-Modelled Transformer for Dynamic Ghost Imaging with Quantum Detectors

ArXiv CS.AI 50%

arXiv:2605.10185v1 Announce Type: cross Abstract: Ghost imaging reconstructs spatial information from a single-pixel bucket detector by correlating structured illumination patterns with scalar intensity measurements. While deep learning approaches have achieved promising results...

<https://arxiv.org/abs/2605.10185>

42 CONF

To Redact, or not to Redact? A Local LLM Approach to Deliberative Process Privilege Classification

ArXiv CS.AI 50%

arXiv:2605.10211v1 Announce Type: cross Abstract: Government transparency laws, like the Freedom of Information (FOIA) acts in the United States and United Kingdom, and the Woo (Open Government Act) in the Netherlands, grant citizens the right to directly request documents from...

<https://arxiv.org/abs/2605.10211>

43 CONF

Generative AI Fuels Solo Entrepreneurship, but Teams Still Lead at the Top

ArXiv CS.AI 50%

arXiv:2605.10291v1 Announce Type: cross Abstract: Recent advances in generative artificial intelligence (AI) are reshaping who enters entrepreneurship, but not who reaches the top of the quality distribution. Using data on over 160,000 product launches on Product Hunt, we find...

<https://arxiv.org/abs/2605.10291>

44 CONF

Qwen Goes Brrr: Off-the-Shelf RAG for Ukrainian Multi-Domain Document Understanding

ArXiv CS.AI 50%

arXiv:2605.10296v1 Announce Type: cross Abstract: We participated in the Fifth UNLP shared task on multi-domain document understanding, where systems must answer Ukrainian multiple-choice questions from PDF collections and localize the supporting document and page. We propose a...

<https://arxiv.org/abs/2605.10296>

45 CONF

Formally Verifying Analog Neural Networks Under Process Variations Using Polynomial Zonotopes

ArXiv CS.AI 50%

arXiv:2605.10474v1 Announce Type: cross Abstract: Analog neural networks are gaining attention due to their efficiency in terms of power consumption and processing speed. However, since analog neural networks are implemented as physical circuits, they are highly sensitive to...

<https://arxiv.org/abs/2605.10474>

46 CONF

iPay: Integrated Payment Action Recognition via Multimodal Networks and Adaptive Spatial Prior Learning

ArXiv CS.AI 50%

arXiv:2605.10732v1 Announce Type: cross Abstract: Automated transit payment analysis is vital for scalable fare auditing and passenger analytics, yet practice still relies on limited manual inspection. Prior vision- and skeleton-based methods remain brittle under noisy onboard...

<https://arxiv.org/abs/2605.10732>

47 CONF

DuetFair: Coupling Inter- and Intra-Subgroup Robustness for Fair Medical Image Segmentation

ArXiv CS.AI 50%

arXiv:2605.10521v1 Announce Type: cross Abstract: Medical image segmentation models can perform unevenly across subgroups. Most existing fairness methods focus on improving average subgroup performance, implicitly treating each subgroup as internally homogeneous. However, this...

<https://arxiv.org/abs/2605.10521>

48 CONF

Acceptance Cards: A Four-Diagnostic Standard for Safe Fine-Tuning Defense Claims

ArXiv CS.AI 50%

arXiv:2605.10575v1 Announce Type: cross Abstract: Safe fine-tuning defenses are often endorsed on the basis of a held-out gap reduction, but the same reduction can come from sampling noise, subject artifacts, capability loss, or a mechanism that does not transfer. We introduce...

<https://arxiv.org/abs/2605.10575>

49 **CONF****Guaranteed Jailbreaking Defense via Disrupt-and-Rectify Smoothing****ArXiv CS.AI** **50%**

arXiv:2605.10582v1 Announce Type: cross Abstract: This paper proposes a guaranteed defense method for large language models (LLMs) to safeguard against jailbreaking attacks. Drawing inspiration from the denoised-smoothing approach in the adversarial defense domain, we propose a...

<https://arxiv.org/abs/2605.10582>

50 **CONF****Fairness vs Performance: Characterizing the Pareto Frontier of Algorithmic Decision Systems****ArXiv CS.AI** **50%**

arXiv:2605.10604v1 Announce Type: cross Abstract: Designing fair algorithmic decision systems requires balancing model performance with fairness toward affected individuals: More fairness might require sacrificing some performance and vice versa, yet the space of possible...

<https://arxiv.org/abs/2605.10604>

51 **CONF****Measuring Embedding Sensitivity to Authorial Style in French: Comparing Literary Texts with Language Model Rewritings****ArXiv CS.AI** **50%**

arXiv:2605.10606v1 Announce Type: cross Abstract: Large language models (LLMs) can convincingly imitate human writing styles, yet it remains unclear how much stylistic information is encoded in embeddings from any language model and retained after LLM rewriting. We investigate...

<https://arxiv.org/abs/2605.10606>

52 **CONF****Re-Trigging Safeguards within LLMs for Jailbreak Detection****ArXiv CS.AI** **50%**

arXiv:2605.10611v1 Announce Type: cross Abstract: This paper proposes a jailbreaking prompt detection method for large language models (LLMs) to defend against jailbreak attacks. Although recent LLMs are equipped with built-in safeguards, it remains possible to craft...

<https://arxiv.org/abs/2605.10611>

53 **CONF****Intrinsic Guardrails: How Semantic Geometry of Personality Interacts with Emergent Misalignment in LLMs****ArXiv CS.AI** **50%**

arXiv:2605.10633v1 Announce Type: cross Abstract: Fine-tuning Large Language Models (LLMs) on benign narrow data can sometimes induce broad harmful behaviors, a vulnerability termed emergent misalignment (EM). While prior work links these failures to specific directions in the...

<https://arxiv.org/abs/2605.10633>

54 **CONF****Towards Understanding Continual Factual Knowledge Acquisition of Language Models: From Theory to Algorithm****ArXiv CS.AI** **50%**

arXiv:2605.10640v1 Announce Type: cross Abstract: Continual Pre-Training (CPT) is essential for enabling Language Models (LMs) to integrate new knowledge without erasing old. While classical CPT techniques like data replay have become the standard paradigm, the mechanisms...

<https://arxiv.org/abs/2605.10640>

55 CONF

bViT: Investigating Single-Block Recurrence in Vision Transformers for Image Recognition

ArXiv CS.AI 50%

arXiv:2605.10661v1 Announce Type: cross Abstract: Vision Transformers (ViTs) are built by stacking independently parameterized blocks, but it remains unclear how much of this depth requires layer specific transformations and how much can be realized through recurrent...

<https://arxiv.org/abs/2605.10661>

56 CONF

Why Low-Resource NLP Needs More Than Cross-Lingual Transfer: Lessons Learned from Luxembourgish

ArXiv CS.AI 50%

arXiv:2605.10714v1 Announce Type: cross Abstract: Cross-lingual transfer has become a central paradigm for extending natural language processing (NLP) technologies to low-resource languages. By leveraging supervision from high-resource languages, multilingual language models can...

<https://arxiv.org/abs/2605.10714>

57 CONF

An Uncertainty-Aware Resilience Micro-Agent for Causal Observability in the Computing Continuum

ArXiv CS.AI 50%

arXiv:2605.10718v1 Announce Type: cross Abstract: Grey failures in the computing continuum produce ambiguous overlapping symptoms that existing approaches fail to diagnose reliably, either due to a lack of causal awareness or acting under high epistemic uncertainty, risking...

<https://arxiv.org/abs/2605.10718>

58 CONF

Break the Brake, Not the Wheel: Untargeted Jailbreak via Entropy Maximization

ArXiv CS.AI 50%

arXiv:2605.10764v1 Announce Type: cross Abstract: Recent studies show that gradient-based universal image jailbreaks on vision-language models (VLMs) exhibit little or no cross-model transferability, casting doubt on the feasibility of transferable multimodal jailbreaks. We...

<https://arxiv.org/abs/2605.10764>

59 CONF

MPerS: Dynamic MLLM MixExperts Perception-Guided Remote Sensing Scene Segmentation

ArXiv CS.AI 50%

arXiv:2605.10769v1 Announce Type: cross Abstract: The multimodal fusion of images and scene captions has been extensively explored and applied in various fields. However, when dealing with complex remote sensing (RS) scenes, existing studies have predominantly concentrated on...

<https://arxiv.org/abs/2605.10769>

60 CONF

The Last Word Often Wins: A Format Confound in Chain-of-Thought Corruption Studies

ArXiv CS.AI 50%

arXiv:2605.10799v1 Announce Type: cross Abstract: Corruption studies, the primary tool for evaluating chain-of-thought (CoT) faithfulness, identify which chain positions are "computationally important" by measuring accuracy when steps are replaced with errors. We identify a...

<https://arxiv.org/abs/2605.10799>

61 CONF

SLIM: Sparse Latent Steering for Interpretable and Property-Directed LLM-Based Molecular Editing

ArXiv CS.AI 50%

arXiv:2605.10831v1 Announce Type: cross Abstract: Large language models possess strong chemical reasoning capabilities, making them effective molecular editors. However, property-relevant information is implicitly entangled across their dense hidden states, providing no explicit...

<https://arxiv.org/abs/2605.10831>

62 CONF

Confidence-Guided Diffusion Augmentation for Enhanced Bangla Compound Character Recognition

ArXiv CS.AI 50%

arXiv:2605.10916v1 Announce Type: cross Abstract: Recognition of handwritten Bangla compound characters remains a challenging problem due to complex character structures, large intra-class variation, and limited availability of high-quality annotated data. Existing Bangla...

<https://arxiv.org/abs/2605.10916>

63 CONF

Polymath: A Challenging Multi-modal Mathematical Reasoning Benchmark

ArXiv CS.AI 50%

arXiv:2410.14702v2 Announce Type: replace Abstract: Multi-modal Large Language Models (MLLMs) exhibit impressive problem-solving abilities in various domains, but their visual comprehension and abstract reasoning skills remain under-evaluated. To this end, we present PolyMATH, a...

<https://arxiv.org/abs/2410.14702>

64 CONF

Leveraging LLMs to Automate Energy-Aware Refactoring of Parallel Scientific Codes

ArXiv CS.AI 50%

arXiv:2505.02184v3 Announce Type: replace Abstract: Large language models (LLMs) are increasingly used for generating parallel scientific codes, with a primary focus on generating functionally correct code. Recent work has focused on generating performant code, with an emphasis...

<https://arxiv.org/abs/2505.02184>

65 CONF

Overcoming Multi-step Complexity in Multimodal Theory-of-Mind Reasoning: A Scalable Bayesian Planner

ArXiv CS.AI 50%

arXiv:2506.01301v2 Announce Type: replace Abstract: Theory-of-Mind (ToM) enables humans to infer mental states-such as beliefs, desires, and intentions-forming the foundation of social cognition. However, existing computational ToM methods rely on structured workflows with...

<https://arxiv.org/abs/2506.01301>

66 CONF

Your Recourse, My Loss? Algorithmic Recourse under Shared Constraints

ArXiv CS.AI 50%

arXiv:2508.11070v2 Announce Type: replace Abstract: Decision makers are increasingly relying on machine learning in sensitive situations. Algorithmic recourse aims to provide individuals with actionable and minimally costly steps to reverse unfavorable AI-driven decisions. While...

<https://arxiv.org/abs/2508.11070>

67 CONF

Recursive Language Models

ArXiv CS.AI 50%

arXiv:2512.24601v3 Announce Type: replace Abstract: We study allowing large language models (LLMs) to process arbitrarily long prompts through the lens of inference-time scaling. We propose Recursive Language Models (RLMs), a general inference paradigm that treats long prompts...

<https://arxiv.org/abs/2512.24601>

68 CONF

Batch-of-Thought: Cross-Instance Learning for Enhanced LLM Reasoning

ArXiv CS.AI 50%

arXiv:2601.02950v3 Announce Type: replace Abstract: Current Large Language Model reasoning systems process queries independently, discarding valuable cross-instance signals such as shared reasoning patterns and consistency constraints. We introduce Batch-of-Thought (BoT), a...

<https://arxiv.org/abs/2601.02950>

69 CONF

Concise Geometric Description as a Bridge: Unleashing the Potential of LLM for Plane Geometry Problem Solving

ArXiv CS.AI 50%

arXiv:2601.21164v3 Announce Type: replace Abstract: Plane Geometry Problem Solving (PGPS) is a multimodal reasoning task that aims to solve a plane geometric problem based on a geometric diagram and problem textual descriptions. Although Large Language Models (LLMs) possess...

<https://arxiv.org/abs/2601.21164>

70 CONF

SayNext-Bench: Why Do LLMs Struggle with Next-Utterance Anticipation?

ArXiv CS.AI 50%

arXiv:2602.00327v2 Announce Type: replace Abstract: We explore the use of large language models (LLMs) for next-utterance anticipation in human dialogue. Despite recent advances in LLMs demonstrating their ability to engage in natural conversations with users, we show that even...

<https://arxiv.org/abs/2602.00327>

71 CONF

TodyComm: Task-Oriented Dynamic Communication for Multi-Round LLM-based Multi-Agent System

ArXiv CS.AI 50%

arXiv:2602.03688v2 Announce Type: replace Abstract: Multi-round LLM-based multi-agent systems rely on effective communication structures to support collaboration across rounds. However, most existing methods employ a fixed communication topology during inference, which falls...

<https://arxiv.org/abs/2602.03688>

72 CONF

When Agents Say One Thing and Do Another: Validating Elicited Beliefs from LLMs

ArXiv CS.AI 50%

arXiv:2602.06286v2 Announce Type: replace Abstract: Large language models (LLMs) are increasingly deployed in high-stakes settings where good decisions require forming beliefs over the probability of unknown outcomes. However, it is unclear whether LLMs act as if they hold...

<https://arxiv.org/abs/2602.06286>

73 CONF

HyPER: Bridging Exploration and Exploitation for Scalable LLM Reasoning with Hypothesis Path Expansion and Reduction

ArXiv CS.AI 50%

arXiv:2602.06527v2 Announce Type: replace Abstract: Scaling test-time compute with multi-path chain-of-thought improves reasoning accuracy, but its effectiveness depends critically on the exploration-exploitation trade-off. Existing approaches address this trade-off in rigid...

<https://arxiv.org/abs/2602.06527>

74 CONF

Interactive Benchmarks

ArXiv CS.AI 50%

arXiv:2603.04737v2 Announce Type: replace Abstract: Existing reasoning evaluation paradigms suffer from different limitations: fixed benchmarks are increasingly saturated and vulnerable to contamination, while preference-based evaluations rely on subjective judgments. We argue...

<https://arxiv.org/abs/2603.04737>

75 CONF

MESD: A Risk-Sensitive Metric for Explanation Fairness Across Intersectional Subgroups

ArXiv CS.AI 50%

arXiv:2603.13452v2 Announce Type: replace Abstract: Fairness in machine learning is predominantly evaluated through outcome-oriented metrics, such as Demographic parity, which measure whether predictions are statistically consistent across protected groups. However, these...

<https://arxiv.org/abs/2603.13452>

76 CONF

PHMForge: Evaluating LLM Agents on Industrial Prognostics through MCP-Native, Algorithm-Grounded Tools

ArXiv CS.AI 50%

arXiv:2604.01532v2 Announce Type: replace Abstract: LLM agents are beginning to invoke industrial asset-management tools through the Model Context Protocol (MCP), yet whether they can act reliably on this substrate for safety-critical \emph{Prognostics and Health Management}...

<https://arxiv.org/abs/2604.01532>

77 CONF

Towards Shutdownable Agents: Generalizing Stochastic Choice in RL Agents and LLMs

ArXiv CS.AI 50%

arXiv:2604.17502v2 Announce Type: replace Abstract: Misaligned artificial agents might resist shutdown. One proposed solution is to train agents to lack preferences between different-length trajectories. The Discounted Reward for Same-Length Trajectories (DReST) reward function...

<https://arxiv.org/abs/2604.17502>

78 CONF

OpenCLAW-P2P v7.0-P2PCLAW: Resilient Multi-Layer Persistence, Live Reference Verification, and Production-Scale Evaluation of Decentralized AI Peer Review v7.0 -- Mathematical Corrections & Ecosystem Developments Edition

ArXiv CS.AI 50%

arXiv:2604.19792v2 Announce Type: replace Abstract: This paper presents OpenCLAW-P2P v7.0, a comprehensive evolution of the decentralized collective-intelligence platform in which autonomous AI agents publish, peer-review, score, and iteratively improve scientific research...

<https://arxiv.org/abs/2604.19792>

79 CONF

Can RL Teach Long-Horizon Reasoning to LLMs? Expressiveness Is Key

ArXiv CS.AI 50%

arXiv:2605.06638v2 Announce Type: replace Abstract: Reinforcement learning (RL) has been applied to improve large language model (LLM) reasoning, yet the systematic study of how training scales with task difficulty has been hampered by the lack of controlled, scalable...

<https://arxiv.org/abs/2605.06638>

80 CONF

Extracting Search Trees from LLM Reasoning Traces Reveals Myopic Planning

ArXiv CS.AI 50%

arXiv:2605.06840v2 Announce Type: replace Abstract: Large language models (LLMs), especially reasoning models, generate extended chain-of-thought (CoT) reasoning that often contains explicit deliberation over future outcomes. Yet whether this deliberation constitutes genuine...

<https://arxiv.org/abs/2605.06840>

81 CONF

Virtual Personas for Language Models via an Anthology of Backstories

ArXiv CS.AI 50%

arXiv:2407.06576v4 Announce Type: replace-cross Abstract: Large language models (LLMs) are trained from vast repositories of text authored by millions of distinct authors, reflecting an enormous diversity of human traits. While these models bear the potential to be used as...

<https://arxiv.org/abs/2407.06576>

82 CONF

Beta Sampling is All You Need: Efficient Image Generation Strategy for Diffusion Models using Stepwise Spectral Analysis

ArXiv CS.AI 50%

arXiv:2407.12173v2 Announce Type: replace-cross Abstract: Generative diffusion models have emerged as a powerful tool for high-quality image synthesis, yet their iterative nature demands significant computational resources. This paper proposes an efficient time step sampling...

<https://arxiv.org/abs/2407.12173>

83 CONF

Capacity-Aware Inference: Mitigating the Straggler Effect in Mixture of Experts

ArXiv CS.AI 50%

arXiv:2503.05066v5 Announce Type: replace-cross Abstract: The Mixture of Experts (MoE) is an effective architecture for scaling large language models by leveraging sparse expert activation to balance performance and efficiency. However, under expert parallelism, MoE suffers from...

<https://arxiv.org/abs/2503.05066>

84 CONF

LLM-FE: Automated Feature Engineering for Tabular Data with LLMs as Evolutionary Optimizers

ArXiv CS.AI 50%

arXiv:2503.14434v3 Announce Type: replace-cross Abstract: Automated feature engineering plays a critical role in improving predictive model performance for tabular learning tasks. Traditional automated feature engineering methods are limited by their reliance on pre-defined...

<https://arxiv.org/abs/2503.14434>

85 CONF

How Instruction and Reasoning Data shape Post-Training: Data Quality through the Lens of Layer-wise Gradients

ArXiv CS.AI 50%

arXiv:2504.10766v2 Announce Type: replace-cross Abstract: As the post-training of large language models (LLMs) advances from instruction-following to complex reasoning tasks, understanding how different data affect finetuning dynamics remains largely unexplored. In this paper,...

<https://arxiv.org/abs/2504.10766>

86 CONF

Elite Polarization in European Parliamentary Speeches: a Novel Measurement Approach Using Large Language Models

ArXiv CS.AI 50%

arXiv:2507.06658v2 Announce Type: replace-cross Abstract: Theories of democratic stability, populism, and party-system crisis often point to a form of polarization that comparative research rarely measures directly: hostile relations among political elites. Existing comparative...

<https://arxiv.org/abs/2507.06658>

87 CONF

Position: Stop Evaluating AI with Human Tests, Develop Principled, AI-specific Tests instead

ArXiv CS.AI 50%

arXiv:2507.23009v2 Announce Type: replace-cross Abstract: Large Language Models (LLMs) have achieved remarkable results on a range of standardized tests originally designed to assess human cognitive and psychological traits, such as intelligence and personality. While these...

<https://arxiv.org/abs/2507.23009>

88 CONF

Assessing Trustworthiness of AI Training Dataset using Subjective Logic -- A Use Case on Bias

ArXiv CS.AI 50%

arXiv:2508.13813v2 Announce Type: replace-cross Abstract: As AI systems increasingly rely on training data, assessing dataset trustworthiness has become critical, particularly for properties like fairness or bias that emerge at the dataset level. Prior work has used Subjective...

<https://arxiv.org/abs/2508.13813>

89 CONF

RL Fine-Tuning Heals OOD Forgetting in SFT

ArXiv CS.AI 50%

arXiv:2509.12235v3 Announce Type: replace-cross Abstract: Supervised Fine-Tuning (SFT) followed by Reinforcement Learning (RL) is a standard post-training recipe for improving Large Language Models (LLM) reasoning, but why it works remains unclear. We revisit the common claim...

<https://arxiv.org/abs/2509.12235>

90 CONF

EverydayMMQA: A Multilingual and Multimodal Framework for Culturally Grounded Spoken Visual QA

ArXiv CS.AI 50%

arXiv:2510.06371v2 Announce Type: replace-cross Abstract: Large-scale multimodal models achieve strong results on tasks like Visual Question Answering (VQA), but they are often limited when queries require cultural and visual information, everyday knowledge, particularly in...

<https://arxiv.org/abs/2510.06371>

91 CONF

Users as Annotators: LLM Preference Learning from Comparison Mode

ArXiv CS.AI 50%

arXiv:2510.13830v2 Announce Type: replace-cross Abstract: Pairwise preference data have played an important role in the alignment of large language models (LLMs). Each sample of such data consists of a prompt, two different responses to the prompt, and a binary label indicating...

<https://arxiv.org/abs/2510.13830>

92 CONF

COGNOS: Universal Enhancement for Time Series Anomaly Detection via Constrained Gaussian-Noise Optimization and Smoothing

ArXiv CS.AI 50%

arXiv:2511.06894v3 Announce Type: replace-cross Abstract: Reconstruction-based methods are a dominant paradigm in time series anomaly detection (TSAD), however, their near-universal reliance on Mean Squared Error (MSE) loss results in statistically flawed reconstruction...

<https://arxiv.org/abs/2511.06894>

93 CONF

Parallel Multi-Circuit Quantum Feature Fusion in Hybrid Quantum-Classical Convolutional Neural Networks for Breast Tumor Classification

ArXiv CS.AI 50%

arXiv:2512.02066v2 Announce Type: replace-cross Abstract: Quantum machine learning has emerged as a promising approach to improve feature extraction and classification tasks in high-dimensional data domains such as medical imaging. In this work, we present a hybrid...

<https://arxiv.org/abs/2512.02066>

94 CONF

When Tables Leak: Attacking String Memorization in LLM-Based Tabular Data Generation

ArXiv CS.AI 50%

arXiv:2512.08875v2 Announce Type: replace-cross Abstract: Large Language Models (LLMs) have recently demonstrated remarkable performance in generating high-quality tabular synthetic data. In practice, two primary approaches have emerged for adapting LLMs to tabular data...

<https://arxiv.org/abs/2512.08875>

95 CONF

Can LLMs Estimate Student Struggles? Human-AI Difficulty Alignment with Proficiency Simulation for Item Difficulty Prediction

ArXiv CS.AI 50%

arXiv:2512.18880v2 Announce Type: replace-cross Abstract: Accurate estimation of item (question or task) difficulty is critical for educational assessment but suffers from the cold start problem. While Large Language Models demonstrate superhuman problem-solving capabilities, it...

<https://arxiv.org/abs/2512.18880>

96 CONF

LENS: LLM-Enabled Narrative Synthesis for Mental Health by Aligning Multimodal Sensing with Language Models

ArXiv CS.AI 50%

arXiv:2512.23025v2 Announce Type: replace-cross Abstract: Multimodal health sensing offers rich behavioral signals for assessing mental health, yet translating these numerical time-series measurements into natural language remains challenging. Current LLMs cannot natively ingest...

<https://arxiv.org/abs/2512.23025>

97 CONF

Big AI is accelerating the metacrisis: What can we do?

ArXiv CS.AI 50%

arXiv:2512.24863v2 Announce Type: replace-cross Abstract: The world is in the grip of ecological, meaning, and language crises that are converging into a metacrisis. Big AI is accelerating them all. LLM engineering sits at the core. Despite the public good motives of language...

<https://arxiv.org/abs/2512.24863>

98 CONF

IntroLM: Introspective Language Models via Prefilling-Time Self-Evaluation

ArXiv CS.AI 50%

arXiv:2601.03511v2 Announce Type: replace-cross Abstract: A major challenge for the operation of large language models (LLMs) is how to predict whether a specific LLM will produce sufficiently high-quality output for a given query. Existing approaches rely on external...

<https://arxiv.org/abs/2601.03511>

99 CONF

What's the plan? Metrics for implicit planning in LLMs and their application to rhyme generation and question answering

ArXiv CS.AI 50%

arXiv:2601.20164v2 Announce Type: replace-cross Abstract: Prior work suggests that language models, while trained on next token prediction, show implicit planning behavior: they may select the next token in preparation to a predicted future token, such as a likely rhyming word,...

<https://arxiv.org/abs/2601.20164>

00 CONF

Training Reasoning Models on Saturated Problems via Failure-Prefix Conditioning

ArXiv CS.AI 50%

arXiv:2601.20829v2 Announce Type: replace-cross Abstract: As Reinforcement Learning with Verifiable Rewards (RLVR) substantially improves the reasoning abilities of large language models (LLMs), a new bottleneck emerges: more training problems become saturated, that is, the LLM...

<https://arxiv.org/abs/2601.20829>

01 CONF

Hidden Heroes and Gradient Bloats: Layer-Wise Redundancy Inverts Attribution in Transformers

ArXiv CS.AI 50%

arXiv:2602.01442v3 Announce Type: replace-cross Abstract: Gradient-based attribution is the workhorse of mechanistic interpretability, yet whether it reliably tracks causal importance at the component level remains largely untested. We causally evaluate this assumption across...

<https://arxiv.org/abs/2602.01442>

02 CONF

Functional Subspace, where language models can use vector algebra to solve problems

ArXiv CS.AI 50%

arXiv:2602.01687v2 Announce Type: replace-cross Abstract: Large language models (LLMs) were invented for natural language tasks such as translation, but they have proved that they can perform highly complex functions across domains. Additionally, they have been thought to...

<https://arxiv.org/abs/2602.01687>

03 CONF

What Structural Inductive Bias Helps Transformers Reason Over Knowledge Graphs? A Study with Tabula RASA

ArXiv CS.AI 50%

arXiv:2602.02834v3 Announce Type: replace-cross Abstract: What structural inductive bias helps transformers reason over knowledge graphs? Through controlled ablations of a minimal transformer modification with four independently removable components (sparse adjacency masking,...

<https://arxiv.org/abs/2602.02834>

04 CONF

BONSAI: Bayesian Optimization with Natural Simplicity and Interpretability

ArXiv CS.AI 50%

arXiv:2602.07144v2 Announce Type: replace-cross Abstract: Bayesian optimization (BO) is a popular technique for sample-efficient optimization of black-box functions. In many applications, the parameters being tuned come with a carefully engineered default configuration, and...

<https://arxiv.org/abs/2602.07144>

05 CONF

SnareNet: Flexible Repair Layers for Neural Networks with Hard Constraints

ArXiv CS.AI 50%

arXiv:2602.09317v2 Announce Type: replace-cross Abstract: Neural networks are increasingly used as fast surrogate models across various domains, but unconstrained predictions can violate physical, operational, or safety requirements. We propose SnareNet, a feasibility-controlled...

<https://arxiv.org/abs/2602.09317>

06 CONF

EcoGym: Evaluating LLMs for Long-Horizon Plan-and-Execute in Interactive Economies

ArXiv CS.AI 50%

arXiv:2602.09514v3 Announce Type: replace-cross Abstract: Long-horizon planning is widely recognized as a core capability of autonomous LLM-based agents; however, current evaluation frameworks suffer from being largely episodic, domain-specific, or insufficiently grounded in...

<https://arxiv.org/abs/2602.09514>

07 CONF

Exploring the AI Obedience: Why is Generating a Pure Color Image Harder than CyberPunk?

ArXiv CS.AI 50%

arXiv:2603.00166v2 Announce Type: replace-cross Abstract: Recent advances in generative AI have shown human-level performance in complex content creation. However, we identify a "Paradox of Simplicity": models that can render complex scenes often fail at trivial, low-entropy...

<https://arxiv.org/abs/2603.00166>

08 CONF

ITLC at SemEval-2026 Task 11: Normalization and Deterministic Parsing for Formal Reasoning in LLMs

ArXiv CS.AI 50%

arXiv:2603.02676v2 Announce Type: replace-cross Abstract: Large language models suffer from content effects in reasoning tasks, particularly in multi-lingual contexts. We introduce a novel method that reduces these biases through explicit structural abstraction that transforms...

<https://arxiv.org/abs/2603.02676>

09 CONF

Relational In-Context Learning via Synthetic Pre-training with Structural Prior

ArXiv CS.AI 50%

arXiv:2603.03805v4 Announce Type: replace-cross Abstract: Relational Databases (RDBs) are the backbone of modern business, yet they lack foundation models comparable to those in text or vision. A key obstacle is that high-quality RDBs are private, scarce and structurally...

<https://arxiv.org/abs/2603.03805>

10 CONF

Generative Experiences for Digital Mental Health Interventions: Evidence from a Randomized Study

ArXiv CS.AI 50%

arXiv:2604.07558v2 Announce Type: replace-cross Abstract: Digital mental health (DMH) tools have extensively explored personalization of interventions to users' needs and contexts. However, this personalization often targets what support is provided, not how it is experienced....

<https://arxiv.org/abs/2604.07558>

11 CONF

SCORP: Scene-Consistent Multi-agent Diffusion Planning with Stable Online Reinforcement Post-Training for Cooperative Driving

ArXiv CS.AI 50%

arXiv:2604.11734v3 Announce Type: replace-cross Abstract: Cooperative driving is a safety- and efficiency-critical task that requires the coordination of diverse, interaction-realistic multi-agent trajectories. Although existing diffusion-based methods can capture multimodal...

<https://arxiv.org/abs/2604.11734>

12 CONF

A Nonasymptotic Theory of Gain-Dependent Error Dynamics in Behavior Cloning

ArXiv CS.AI 50%

arXiv:2604.14484v2 Announce Type: replace-cross Abstract: Behavior cloning (BC) policies on position-controlled robots inherit the closed-loop response of the underlying PD controller, yet the nonasymptotic finite-horizon consequences of controller gains for BC failure remain...

<https://arxiv.org/abs/2604.14484>

13 CONF

Benchmarking Real-Time Question Answering via Executable Code Workflows

ArXiv CS.AI 50%

arXiv:2604.16349v2 Announce Type: replace-cross Abstract: Retrieving real-time information is a fundamental capability for search-integrated agents in real-world applications. However, existing benchmarks are predominantly static and therefore fail to capture the temporal...

<https://arxiv.org/abs/2604.16349>

14 CONF

enclawed: A Configurable, Sector-Neutral Hardening Framework for Single-User AI Assistant Gateways

ArXiv CS.AI 50%

arXiv:2604.16838v2 Announce Type: replace-cross Abstract: We present enclawed, a hard-fork hardening framework built on the OpenClaw AI assistant gateway. enclawed targets deployments that need attestable peer trust, deny-by-default external connectivity, signed-module loading,...

<https://arxiv.org/abs/2604.16838>

15 CONF

Generative Synthetic Data for Causal Inference: Pitfalls, Remedies, and Opportunities

ArXiv CS.AI 50%

arXiv:2604.23904v2 Announce Type: replace-cross Abstract: Synthetic tabular data are often evaluated by distributional similarity, privacy distance, or train-on-synthetic-test-on-real predictive performance, but these criteria do not ensure validity for causal inference. We show...

<https://arxiv.org/abs/2604.23904>

16 CONF

Static Program Slicing Using Language Models With Dataflow-Aware Pretraining and Constrained Decoding

ArXiv CS.AI 50%

arXiv:2604.26961v2 Announce Type: replace-cross Abstract: Static program slicing is a fundamental software engineering technique for isolating code relevant to specific variables. While recent learning-based approaches using language models (LMs) show promise in automating slice...

<https://arxiv.org/abs/2604.26961>

17 CONF

The Perceptual Bandwidth Bottleneck in Vision-Language Models: Active Visual Reasoning via Sequential Experimental Design

ArXiv CS.AI 50%

arXiv:2605.01345v3 Announce Type: replace-cross Abstract: Visual perception in modern Vision-Language Models (VLMs) is constrained by a perceptual bandwidth bottleneck: a broad field of view preserves global context but sacrifices the fine-grained details required for complex...

<https://arxiv.org/abs/2605.01345>

18 CONF

PersonaTeaming: Supporting Persona-Driven Red-Teaming for Generative AI

ArXiv CS.AI 50%

arXiv:2605.05682v2 Announce Type: replace-cross Abstract: Recent developments in AI safety research have called for red-teaming methods that effectively surface potential risks posed by generative AI models, with growing emphasis on how red-teamers' backgrounds and perspectives...

<https://arxiv.org/abs/2605.05682>